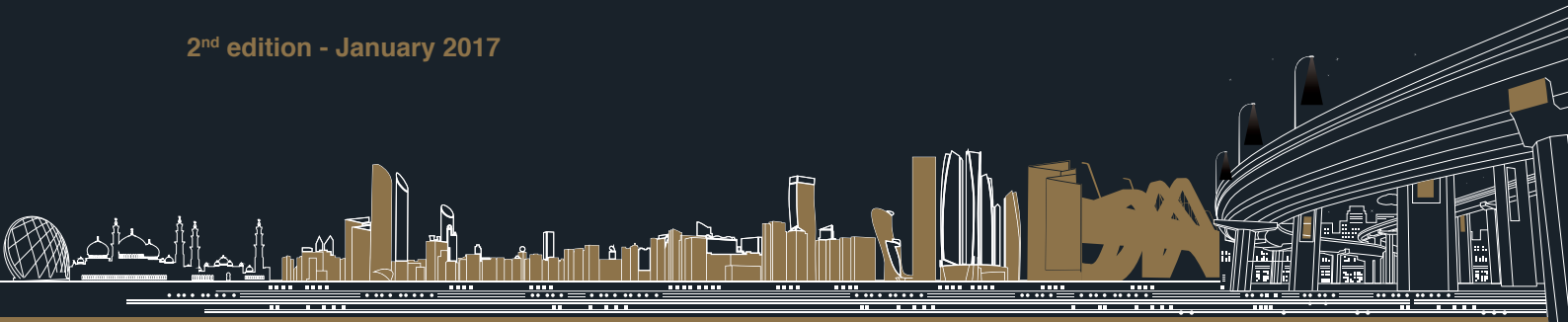


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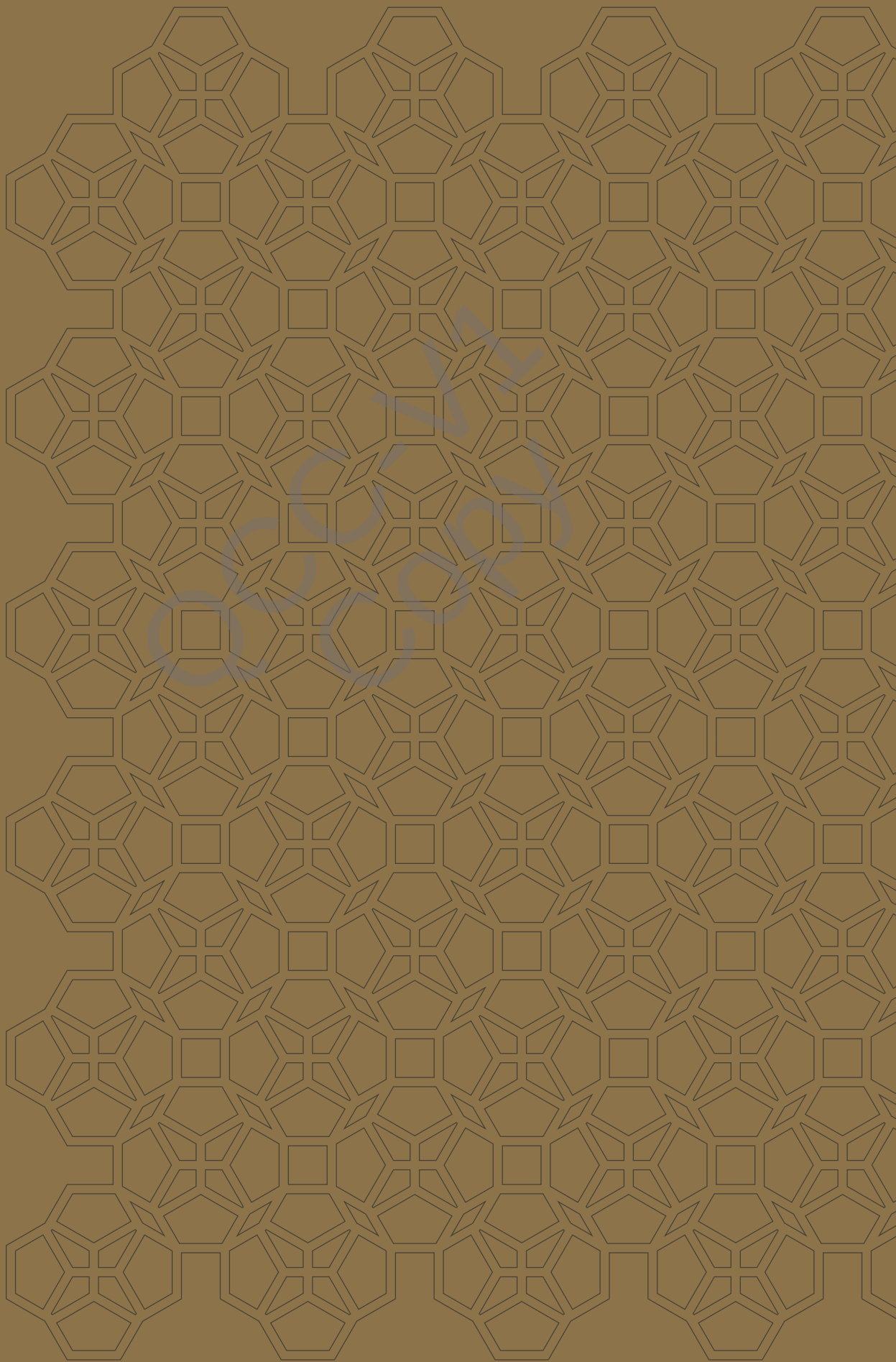
Abu Dhabi Emirate Guideline for Infrastructure Services Standards

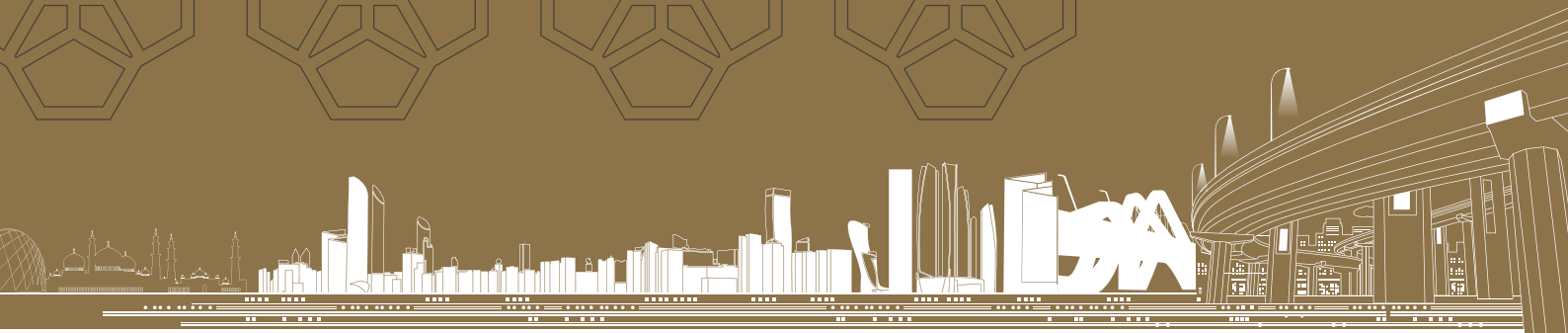
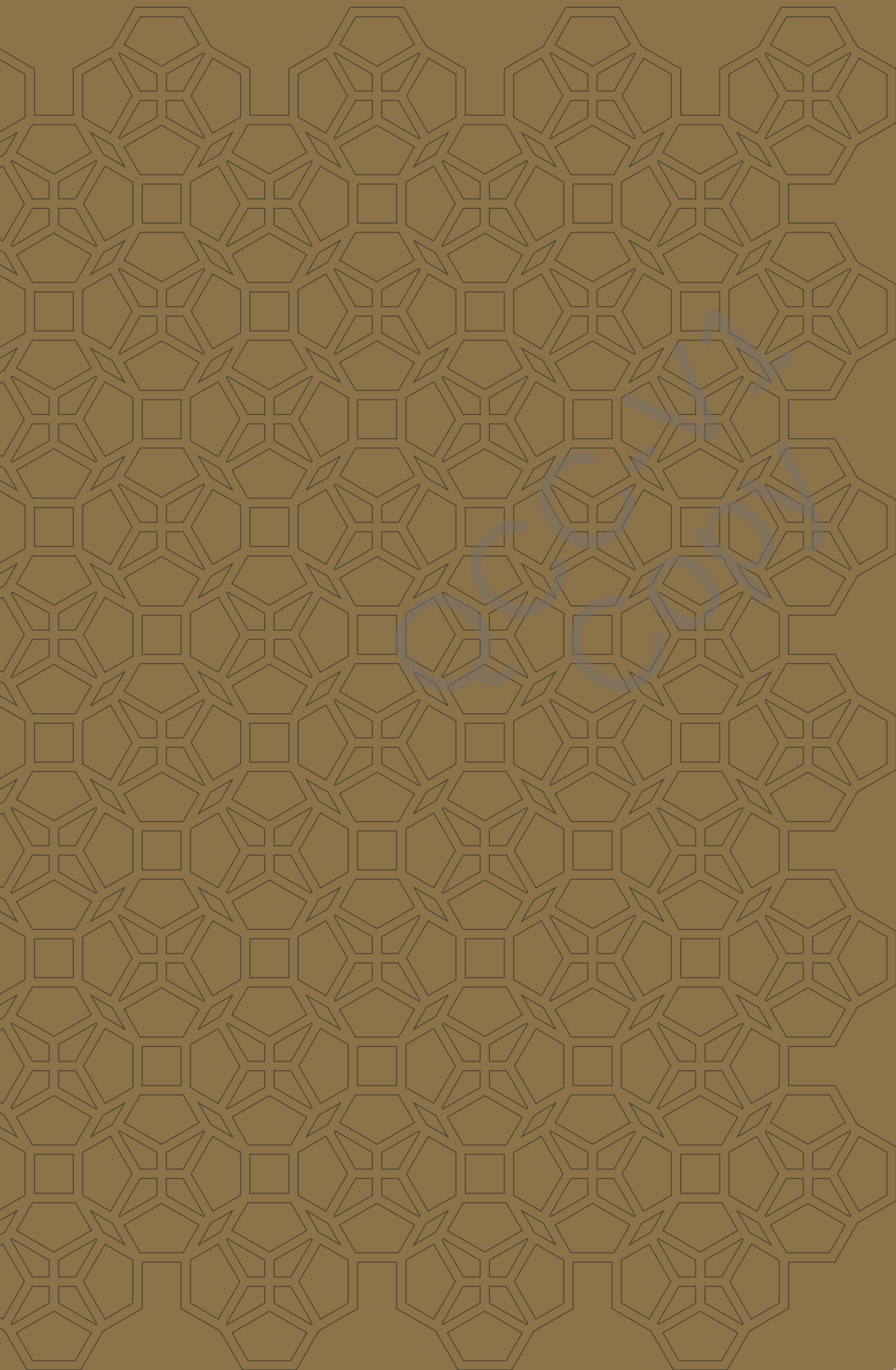
2nd edition - January 2017



مجلس أبوظبي للجودة والمطابقة
ABU DHABI QUALITY AND CONFORMITY COUNCIL









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Plan Al Ain 2030: Al Ain Urban Structure Framework	(DP-302)
Development Review Stream: Master Plan and Project	(DP-303)
Community Facility Planning Standards	(DP-304)
Safety and Security Planning Manual	(DP-305)
Estidama Pearl Building Rating System: Design and Construction	(DP-306)
Estidama Pearl Community Rating System: Design and Construction	(DP-307)
Estidama Pearl Villa Rating System: Design and Construction	(DP-308)
Public Realm Design Manual (PRDM)	(PR-401)
Lighting Manual	(PR-402)
Design Manual: Minimum Requirements for Private School Facilities	(PR-403)
Design Manual: Standards and Criteria for School Facilities	(PR-404)
Mosque Design Regulations	(PR-405)
Abu Dhabi Surface Transport Master Plan	(TR-501)
Al Ain Surface Transport Master Plan	(TR-502)
Al Gharbia Surface Transport Master plan	(TR-503)
Eastern Region Surface Transport Master plan	(TR-504)
Freight Master Plan - Truck Parking Standards	(TR-505)
Addressing System Standards Manual (Volume II), Programme Design Policies, Authority and Organization	(TR-506-1)
Addressing System Standards Manual (Volume III) Geonames Standard and Practitioner's User Guide	(TR-506-2)
Addressing System Standards Manual (Volume IV), Street Addressing Standard and Practitioner's User Guide	(TR-506-3)
Addressing System Standards Manual (Volume V), Signage Standard and Practitioner's User Guide	(TR-506-4)
Congestion Management Policy and Procedures	(TR-507)
Environmental Impact Assessment for Road Projects	(TR-508)
Geotechnical Investigation and Design (Volume I) Managing Geotechnical Risk	(TR-509-1)
Geotechnical Investigation and Design (Volume II) Ground Investigation and Geotechnical Design	(TR-509-2)
Land Surveying and Mapping Guide for Road Projects	(TR-510)
Manual on Uniform Traffic Control Devices	(TR-511)
Non-Disruptive Road Crossings for utilities Manual	(TR-512)
Pavement Design Manual	(TR-513)

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Road Geometric Design Manual	(TR-514)
Road Performance Monitoring Systems Guide	(TR-515)
Road Structures Design Manual	(TR-516)
Road user code	(TR-517)
Roadside Design Guide	(TR-518)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Technical circular –Bus Stop Design standard drawings	(TR-520)
Guide Drawings on locating Bus stops on Urban Roads	(TR-521)
Technical Guidance Document for Construction Environmental Management Plan	(TR-522)
Technical Guidance Document for Environmental Audit Report	(TR-523)
Technical Guidance Document for Environmental Impact Assessment	(TR-524)
Technical Guidance Document for Preliminary Environmental Review	(TR-525)
Technical Guidance Document for Strategic Environmental Impact Assessment	(TR-526)
Technical Guidance Document for Submission of Environmental Permit Applications and Reports	(TR-527)
Traffic Calming Guideline	(TR-528)
Traffic Signals and Electronic Warning Information Systems Manual	(TR-529)
Walking and Cycling Master Plan – Network Design	(TR-530)
Work Zone Traffic Management Manual	(TR-531)
Access Management Policy & Procedure	(TR-532)
Bus Rapid Transit Planning Standards	(TR-533)
Bus Services Planning Standards	(TR-534)
Metro Planning Standards	(TR-535)
Railway Planning Standards	(TR-536)
Tramways Planning Standards	(TR-537)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Urban Street Design Manual (USDM)	(ROW-603)
Utility Plot Design Manual (UPDM)	(ROW-604)
Drinking Water Safety Plan	(WA-701)
Hygienic Practices within Pumping Station and Water Network	(WA-702)
Metering and Data Exchange Code	(WA-703)
Standard Technical Specifications for Water Works	(WA-704)



List of Documens	Ref. Code
The Water Quality Regulations	(WA-705)
The Water Transmission Code	(WA-706)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
The Water Supply Regulations	(WA-708)
Guide to Water Supply Regulations	(WA-709)
The Water Distribution Code	(WA-710)
The Customer Metering Regulations	(WA-711)
Guideline for Work in Vicinity of ADDC Assets	(WA-712)
Design Guidelines	(WA-713)
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Technical Data Sheet for Electrical Works	(WA-719)
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Recycled Water and Bio-solids Regulations	(WA-722)
Guide to the Recycled Water and Bio-solids Regulations	(WA-723)
Guide to the Trade Effluent Control Regulations	(WA-724)
Irrigation Manual (Volume I) Design Manual	(WA-725-1)
Irrigation Manual (Volume II) Operation and Maintenance Manual	(WA-725-2)
Irrigation Manual (Volume III) Technical Specifications	(WA-725-3)
Irrigation Manual (Volume IV) Standard Drawings	(WA-725-4)
Stormwater and subsoil drainage systems (Volume I) Design Manual	(WA-726-1)
Stormwater and subsoil drainage systems (Volume II) Standard Drawings	(WA-726-2)
Stormwater and subsoil drainage systems (Volume III)Operation and Maintenance Management Manual	(WA-726-3)
Stormwater and subsoil drainage systems Standard Technical Specification (Volume 1, 2 & 3)	(WA-726-4)
The Electricity Transmission Code	(EN-801)
Electricity Transmission Security Standard	(EN-802)
Standard Technical Specifications for Electricity 400-220-132-33-22-11 kV and Low Voltage Auxiliary System	(EN-803)
Electricity Design Guidelines of ADDC (5) year Planning Statement	(EN-804)

List of Documens	Ref. Code
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)
Electricity Distribution code	(EN-806)
Electricity Wiring Regulation Code	(EN-807)
Engineering Standards for Earthing	(EN-808)
LV Design Guidelines	(EN-809)
Standard Technical Specification for Electrical Distribution Assets for 33KV, 22KV, 11 KV and Low Voltage Systems	(EN-810)
Material and Equipment Standards and Specifications	(GA-901)
Gas Distribution System - Construction Standards and Specifications - Polyethylene Distribution Mains and Gas Services	(GA-902)
Gas Distribution System - Construction Standards and Specifications - Steel Distribution Pipelines and Gas Services	(GA-903)
Design Basis Memorandum	(GA-904)
Telecom Civil Infrastructure Guidelines for Fixed Networks - Design Requirements	(TE-1101)
Telecom Civil Infrastructure Guidelines for Fixed Networks - Implementation and Specifications	(TE-1102)
Common Standards and Specifications on Civil Works for Mobile Network Site Construction	(TE-1103)
Common In-Building Specification Guideline	(TE-1104)
Waste Classification Policy	(WM-1201)
Waste Planning Policy	(WM-1202)
Licensing and Enforcement Policy for Waste Sector	(WM-1203)
Waste Collection, Segregation, Transfer and Tracking Policy	(WM-1204)
Waste Classification Technical Guideline	(WM-1205)
Waste Reuse, Recycling, Resource Recovery, Waste to Energy, Treatment and Disposal Policy	(WM-1206)
Certification Scheme for Recycled Crushed Aggregate	(WM-1207)
Certification Scheme for Re-refined Engine Lubricant Base Oil	(WM-1208)
Certification Scheme for Biodiesel	(WM-1209)
Requirements and Procedures for the Disposal of Hazardous Wastes (CWMTG 2# merged)	(WM-1210)
Requirements and Procedures for Registration of Skips and Container	(WM-1211)
Registration of Waste Management Consultancy Offices	(WM-1212)
Inspection of Waste Treatment Facilities, Waste Transportation and Pest Control Facilities	(WM-1213)
Permits and Licensing Requirements for Transportation, Treatment and Recycling Facility	(WM-1214)
Management of Asbestos / Asbestos Containing Material (ACM)	(WM-1215)





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شكر وتقدير

Acknowledgement



Abu Dhabi Quality & Conformity Council would like to express special thanks of gratitude to all governmental entities and technical committees participated in the development of the Abu Dhabi Emirate Guideline for Infrastructure Services Standards (ISGL).

يتقدم مجلس أبوظبي للجودة والمطابقة بجزيل الشكر والإمتنان لجميع الجهات الحكومية وأعضاء اللجان الفنية لمساهماتهم الفعالة في تطوير الدليل الإرشادي لمواصفات البنية التحتية للخدمات لإمارة أبوظبي.

Main Participants:

الجهات المشاركة:

• Department of Municipal Affairs and Transportation	• دائرة الشؤون البلدية والنقل
• Abu Dhabi Water and Electricity Authority	• هيئة مياه وكهرباء أبوظبي
• Environment Agency – Abu Dhabi	• هيئة البيئة – أبوظبي
• Abu Dhabi Education Council	• مجلس أبوظبي للتعليم
• Abu Dhabi Urban Planning Council	• مجلس أبوظبي للتخطيط العمراني
• Regulation and Supervision Bureau	• مكتب التنظيم والرقابة
• Center of Waste Management (Tadweer)	• مركز إدارة النفايات – تدوير
• Abu Dhabi Municipality	• بلدية مدينة أبوظبي
• Al Ain Municipality	• بلدية مدينة العين
• Western Region Municipality	• بلدية مدينة المنطقة الغربية
• Abu Dhabi Distribution Company	• شركة أبوظبي للتوزيع
• Al Ain Distribution Company	• شركة العين للتوزيع
• Abu Dhabi Transmission & Despatch Company	• شركة أبوظبي للنقل والتحكم (ترانسكو)
• Abu Dhabi Sewerage Services Company	• شركة أبوظبي لخدمات الصرف الصحي

• Abu Dhabi General Services Company	• شركة أبوظبي للخدمات العامة (مساندة)
• Abu Dhabi National Oil Company (ADNOC-Distribution)	• شركة أدنوك للتوزيع
• Abu Dhabi Gas Industries Limited	• شركة أبوظبي لصناعات الغاز المحدودة
• Ministry of Interior (Abu Dhabi Police and Civil Defence)	• وزارة الداخلية (القيادة العامة لشرطة أبوظبي والدفاع المدني)
• Telecommunications Regulatory Authority	• هيئة تنظيم الاتصالات
• United Arab Emirates University	• جامعة الإمارات العربية المتحدة
• Masdar Institute	• معهد مصدر
• Emirates Telecommunications Corporation	• مؤسسة الإمارات للاتصالات (اتصالات)
• Emirates Integrated Telecommunications Company	• شركة الإمارات للاتصالات المتكاملة (دو)
• Al Yah Satellite Communications Company	• شركة الياه سات للاتصالات الفضائية
• Dolphin Energy	• شركة دولفين للطاقة
• Tabreed	• شركة تبريد
• Al Fanar Gas Services	• الفنار لخدمات الغاز
• Union Gas	• الغاز المتحدة





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المقدمة Introduction



2.1 Background

The Abu Dhabi Government entities are seeking to develop appropriate infrastructure while preserving the environment for achieving environmental sustainability in line with the requirements of Abu Dhabi 2030 vision.

In 2011, H.H. Shaikh Hazza Bin Zayed Al Nahyan, under the auspices of the Executive Council, initiated the Infrastructure Standards Evaluation Project (ISEP). One of the main motivations for executing the project was to ensure the cost-effectiveness of infrastructure services. What's more, one of the factors that were prioritized as part of the ISEP is to harmonize the regulatory environment for each element of infrastructure.

In April 2013, the responsibility for the Abu Dhabi Emirate Guideline for Infrastructure Services Standards (ISGL) was transferred to the Abu Dhabi Quality and Conformity Council (ADQCC). Since that time, the ADQCC has identified opportunities to improve cost effectiveness of infrastructure services by the following:

- Harmonizing the standards, guidelines, and regulations across all entities.
- Enabling the acceptance of the (ISGL) by all relevant authorities to ultimately adopt it as a reference for infrastructure services in Abu Dhabi Emirate.

The Abu Dhabi Quality & Conformity Council has contributed to ensure harmonization of the regulatory environment of each element of the infrastructure through the involvement of stakeholders.

1.2 نبذة عن المشروع:

تعمل جهات حكومة أبوظبي على تطوير بنية تحتية مناسبة مع الحفاظ على البيئة وذلك لتحقيق الاستدامة البيئية بما يتسق مع متطلبات رؤية أبوظبي 2030.

ففي عام 2011، أطلق سمو الشيخ هزاع بن زايد آل نهيان مشروع تقييم معايير البنية التحتية (ISEP) وذلك تحت رعاية المجلس التنفيذي الموقر. وكان أحد الدوافع الأساسية لتنفيذ المشروع يكمن في ضمان فعالية تقدير تكاليف خدمات البنية التحتية. كما أن أحد العوامل التي تصدرت الأولويات كجزء من مشروع تقييم معايير البنية التحتية يكمن في تهيئة بيئة مبنية على المواءمة والتناغم لكل عنصر من عناصر البنية التحتية.

وفي شهر أبريل عام 2013، انتقلت المسؤولية عن (الدليل الإرشادي لمواصفات البنية التحتية للخدمات لإمارة أبوظبي) إلى مجلس أبوظبي للجودة والمطابقة. ومنذ ذلك الحين، يقوم مجلس أبوظبي للجودة والمطابقة بتحديد فرص تحسين فعالية تقدير تكلفة خدمات البنية التحتية وذلك من خلال ما يلي:

- ضبط التناغم بين المواصفات والأدلة الإرشادية والتشريعات المتعلقة بها داخل كافة الجهات المعنية.
- تمكين الجهات المعنية من قبول الدليل ليتم تبنيه كمرجع لخدمات البنية التحتية في إمارة أبوظبي.

وقد ساهم مجلس أبوظبي للجودة والمطابقة وما زال يساهم في ضمان التناغم في البيئة التنظيمية لكل عنصر من عناصر البنية التحتية وذلك من خلال التعاون مع الشركاء المعنيين.

2.2 Purpose and Scope

The purpose of Abu Dhabi Emirate (ISGL). Is to provide a comprehensive guide on the various documents that set out the applicable standards and specifications for the Infrastructure services.

This guideline shall serve all government agencies, Consultants, developers, contractors and others who participate in the planning, design, management, operation, and the maintenance and of the infrastructure services.

The (ISGL) is aligned with the Abu Dhabi Vision 2030. It has been developed in coordination with all relevant entities to obtain details of their current practices and procedures for infrastructure development and their Operation and Maintenance standards.

This guideline covers ten different areas of infrastructure services in the Emirate of Abu Dhabi as follows:

- Development Planning.
- Public Realm & Community Facilities Infrastructure
- Transport Infrastructure
- Right of Way
- Integrated Water Management Infrastructure
- Energy Infrastructure
- Gas Infrastructure
- District Cooling Infrastructure
- Telecommunications Infrastructure
- Waste Management Infrastructure

This is a living document that would accept updates and development through the collaboration of partnering entities. What is more, ADQCC is obliged to review it frequently to make sure that the information presented is complete, correct and up to date.

2.2 الهدف ونطاق الدليل

يكمّن الهدف من الدليل في كونه مرجعاً ودليلاً شاملاً حول مختلف الوثائق الفنية التي تشمل المواصفات والمعايير المطبقة والمتعلقة في البنية التحتية.

يخدم هذا الدليل كافة الجهات الحكومية، والاستشاريين، والمطورين، والمقاولين، وغيرهم ممن يشاركون في التخطيط والتصميم وإدارة وتشغيل وصيانة خدمات البنية التحتية.

يتسق الدليل مع رؤية أبوظبي 2030. حيث تم تطويره وتحديثه بالتنسيق مع كافة الجهات المعنية للحصول على تفاصيل الممارسات والإجراءات الحالية التي يتبعونها لتطوير البنية التحتية ومعايير التشغيل والصيانة لديهم.

يغطي هذا الدليل عشرة مجالات مختلفة من مجالات خدمات البنية التحتية داخل إمارة أبوظبي وهي:

- التخطيط التطويري
- خدمات الأماكن العامة والمرافق المجتمعية
- البنية التحتية للنقل
- مسارات حرم الطريق
- البنية التحتية لإدارة المياه المتكاملة
- البنية التحتية للطاقة
- البنية التحتية للغاز
- البنية التحتية لتبريد المناطق
- البنية التحتية للاتصالات
- البنية التحتية لإدارة النفايات

يعتبر هذا الدليل وثيقة حية قابلة للتحديث والتطوير ويلتزم مجلس أبوظبي للجودة والمطابقة بالعمل على مراجعته بشكل دوري ومستمر وذلك بالتعاون مع الجهات المعنية في إمارة أبوظبي للتأكد من تكاملية المعلومات وصحتها وحداثتها.



2.3 Outcomes

This new edition of the (ISGL), is an example of the continual improvement by the ADQCC to better understand infrastructure services, and to accommodate changes from input by all stakeholders.

The text of the new version of (ISGL) offers direction and guidance to the user on the relevant documents for various infrastructure elements, which include the updated version of these documents.

Additionally, the online version of this edition of the (ISGL) at www.qcc.abudhabi.ae provides direct access to the infrastructure standards and specifications through a web-based download. This feature enables more widespread access to information, which is aligned with best practice on transparency.

The 2016 edition of the (ISGL) also reflects changes that were collectively made during the deliberation of the Working Groups.

3.2 المخرجات

يمثل هذا الإصدار الجديد من الدليل نموذجاً للتحسين المتواصل من جانب مجلس أبوظبي للجودة والمطابقة لتحقيق الفهم الأفضل لخدمات البنية التحتية وتلبية التعديلات المنبثقة عن الشركاء المعنيين.

يزود نص الإصدار الجديد من الدليل المستخدم بتوجيهات وإرشادات حول الوثائق المتعلقة بمختلف عناصر البنية التحتية والتي تتضمن نسخة محدثة من هذه الوثائق.

إضافةً إلى ذلك، فإن النسخة الإلكترونية من الدليل والمتوفر على الموقع الإلكتروني للمجلس www.qcc.abudhabi.ae، توفر لمستخدم الدليل فرصة الاطلاع على معايير ومواصفات البنية التحتية من خلال تحميلها مباشرة من على الموقع الإلكتروني. تمكن هذه الخاصية من نشر محتوى الدليل على نطاق أوسع، وهذا الإجراء يتسق مع أفضل ممارسات الشفافية. كما يعكس إصدار 2016 من الدليل التعديلات التي تم التوافق عليها بشكل جماعي من قبل جميع الأعضاء المشاركين في مجموعات العمل الفنية المشاركة في إعداد هذا الدليل.

2.4 Content and Format

The content of this document is structured as follows:

Chapter 1 includes the Acknowledgment.

Chapter 2 provides an introduction and some background on the (ISEP) project.

Chapter 3 is devoted to the Planning processes for infrastructure development.

Chapter 4 describes the infrastructure element of the Public Realm and Community Facilities in terms of key design principles and applicable standards and guidelines.

Chapter 5 presents the applicable standards and guidelines for the Transport Infrastructure.

4.2 المحتوى والصياغة

تم تصميم محتوى هذه الوثيقة كما يلي:

الفصل الأول والذي يتضمن شكر وتقدير.

الفصل الثاني والذي يتضمن نبذة عن مشروع تقييم معايير البنية التحتية (ISEP).

الفصل الثالث وهو مخصص لإجراءات التخطيط لتطوير البنية التحتية.

الفصل الرابع والذي يصف عناصر البنية التحتية لخدمات الأماكن العامة والمرافق المجتمعية فيما يتعلق بمبادئ التصميم الرئيسية والمواصفات والأدلة الإرشادية المطبقة.

الفصل الخامس والذي يعرض المواصفات والأدلة الإرشادية المطبقة بالنسبة للبنية التحتية الخاصة بالنقل.

Chapter 6 presents the applicable standards and guidelines for the Right of Way (ROW) and Utility Disposition.

Chapter 7 presents the applicable standards and guidelines for the Integrated Water Management Infrastructure.

Chapter 8 presents the applicable standards and guidelines for the Energy Infrastructure.

Chapter 9 presents the applicable standards and guidelines for the Gas Infrastructure.

Chapter 10 presents the applicable standards and guidelines for the District Cooling Infrastructure.

Chapter 11 presents the applicable standards and guidelines for the Telecommunications Infrastructure.

Chapter 12 presents the applicable standards and guidelines for the Waste Management Infrastructure.

Chapter 13 deals with the permitting requirements for infrastructure development (NOC) program for utilities and infrastructures in the emirate of Abu Dhabi.

Appendix 2 is a collection of the acronyms used throughout the document, as well as the glossary of key terms with definitions.

Appendix 3 identifies the key entities involved in the Infrastructure services provision process, and includes a brief overview of their respective roles and responsibilities in this regard.

الفصل السادس والذي يعرض المواصفات والأدلة الإرشادية المطبقة لمسارات حرم الطريق (ROW).

الفصل السابع والذي يعرض المواصفات والأدلة الإرشادية المطبقة بالنسبة للبنية التحتية لإدارة المياه المتكاملة.

الفصل الثامن والذي يعرض المواصفات والأدلة الإرشادية المطبقة بالنسبة للبنية التحتية للطاقة.

الفصل التاسع والذي يعرض المواصفات والأدلة الإرشادية المطبقة بالنسبة للبنية التحتية للغاز.

الفصل العاشر والذي يعرض المواصفات والأدلة الإرشادية المطبقة بالنسبة للبنية التحتية لتبريد المناطق.

الفصل الحادي عشر والذي يعرض المواصفات والأدلة الإرشادية المطبقة بالنسبة للبنية التحتية للاتصالات.

الفصل الثاني عشر والذي يعرض المواصفات والأدلة الإرشادية المطبقة بالنسبة للبنية التحتية لإدارة النفايات.

الفصل الثالث عشر ويغطي متطلبات منح التصاريح لتطوير البنية التحتية، وبرنامج شهادات عدم الممانعة (NOC) الخاص بالمرافق والبنية التحتية في إمارة أبوظبي.

الملحق 2 وهو عبارة عن حصر للاختصارات المستخدمة في كامل الوثيقة ومسرد للمصطلحات والتعريفات الرئيسية.

الملحق 3 والذي يحدد الجهات الرئيسية المشاركة في عملية تقديم خدمات البنية التحتية ويتضمن أيضاً نظرة مختصرة لأدوار ومسؤوليات هذه الجهات في هذا الصدد.





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3

DEVELOPMENT **PLANNING**



3.1 Context in Abu Dhabi

Abu Dhabi's rapid development and urbanization is continuously increasing. To ensure alignment with the overarching principles of Abu Dhabi Vision 2030, all developments within the Emirate are subject to rigorous review by the Urban Planning Council.

Abu Dhabi city, the largest city in Abu Dhabi Emirate and capital of the UAE, has developed a tremendous amount since the discovery of oil in the mid-20th century. Now, it has become one of the most developed and renowned cities in the world with the number of developments in Abu Dhabi city continuously increasing in response to population growth and economic development. Table 1 shows the existing and projected population for Abu Dhabi city. These projections form the basis of the city's urban structure framework plan developed by Urban Planning Council (UPC).

Table 1: Population Projections for Abu Dhabi city

Year	Population (millions)
2015 ¹	1.72
2020 ²	1.9
2030 ²	2.4
Notes: 1) Adapted from SCAD Statistical Yearbook 2016. 2) Adapted from Urban Planning Council.	

The Al Ain Region is considered the spiritual heart of Abu Dhabi emirate and a historical place that encompasses the cultural memory and ancient wisdom of the nation. Development planning structures applied to the Al Ain Region differ to that of the capital city. These planning structures aim at providing a high quality of life for the public, while maintaining the ecological and cultural heritage of the region. Table 2 Provides population projections for Al Ain Region. These projections are the basis of the city's urban structure framework developed by UPC.

Table 2: Population Projections for Al Ain Region

Year	Population (thousands)
2015 ¹	739
2020 ²	854
2030 ²	1,086
Notes: 1) Adapted from SCAD Statistical Yearbook 2016. 2) Adapted from Urban Planning Council.	

Al Gharbia, the Western Region of Abu Dhabi emirate, is approximately 60,000 km² which accounts for 83% of the emirate's total area. Al Gharbia has a long term strategy where it will see major expansions and developments within different sectors including education, healthcare, housing, energy, retail, leisure and infrastructure. Table 3 provides the population projections for Al Gharbia.

Table 3: Population projections for Al Gharbia

Year	Population (thousands)
2015 ¹	326
2020 ²	374
2030 ²	490
Notes: 1) Adapted from SCAD Statistical Yearbook 2016. 2) Adapted from Urban Planning Council.	

3.2 Structured Development Policy

The UPC has developed a structured development policy that aims to establish a structured process for the planning, implementation and development of area plans and infill developments. For this to be achieved, the policy outlines the requirement for a coordinated design and phasing of development projects, infrastructure, community facilities and public realm that are compliant and coordinated with future development and the Growth Management Strategy.

A main objective of the policy is to ensure relevant government departments and services providers have created a coordinated phasing plan for all plot types that is legible and realistic and covers all types of land use. In turn, this will help to establish a degree of certainty for the owner when developing a plot during the initial planning and financing activity phase.

The general approach for a development methodology is to package the relevant planning components together (e.g., land use, infrastructure, community facilities, public realm) so that when a detailed plan is prepared for a greenfield site (also known as an area plan or infill site), the development phasing, associated costs and timelines are also known and provided.

The UPC, in association with other government departments and services providers, is tasked with the responsibility to undertake the following planning activities:

- Determine the demand for various land uses based on an assessment of Vision 2030, Economic Vision 2030 and projected growth of key industry sectors, population forecasts, requests for plots, etc.
- Determine the areas of the city that are appropriate for new and infill development and prepare a high level phasing plan;



- Identify appropriate land uses and estimate the number of plots in each area;
- Prepare detailed development plans, as required, for area plans and infill developments; and
- Prepare a phasing plan for each area plan and infill development that includes infrastructure, community facilities and Public Realm along with associated budgets and timeframes.

In terms of the development review process, the UPC is responsible for the planning review and coordination of inter-agency reviews for all strategic development projects across the emirate of Abu Dhabi. Development submissions reviewed by the UPC are categorized based on certain criteria, as detailed in Figure 1. The UPC has two review streams, based on specific criteria, tailored to meet the needs of each development type. The UPC refers to the Planning Review process as Development Review Stream (DRS).

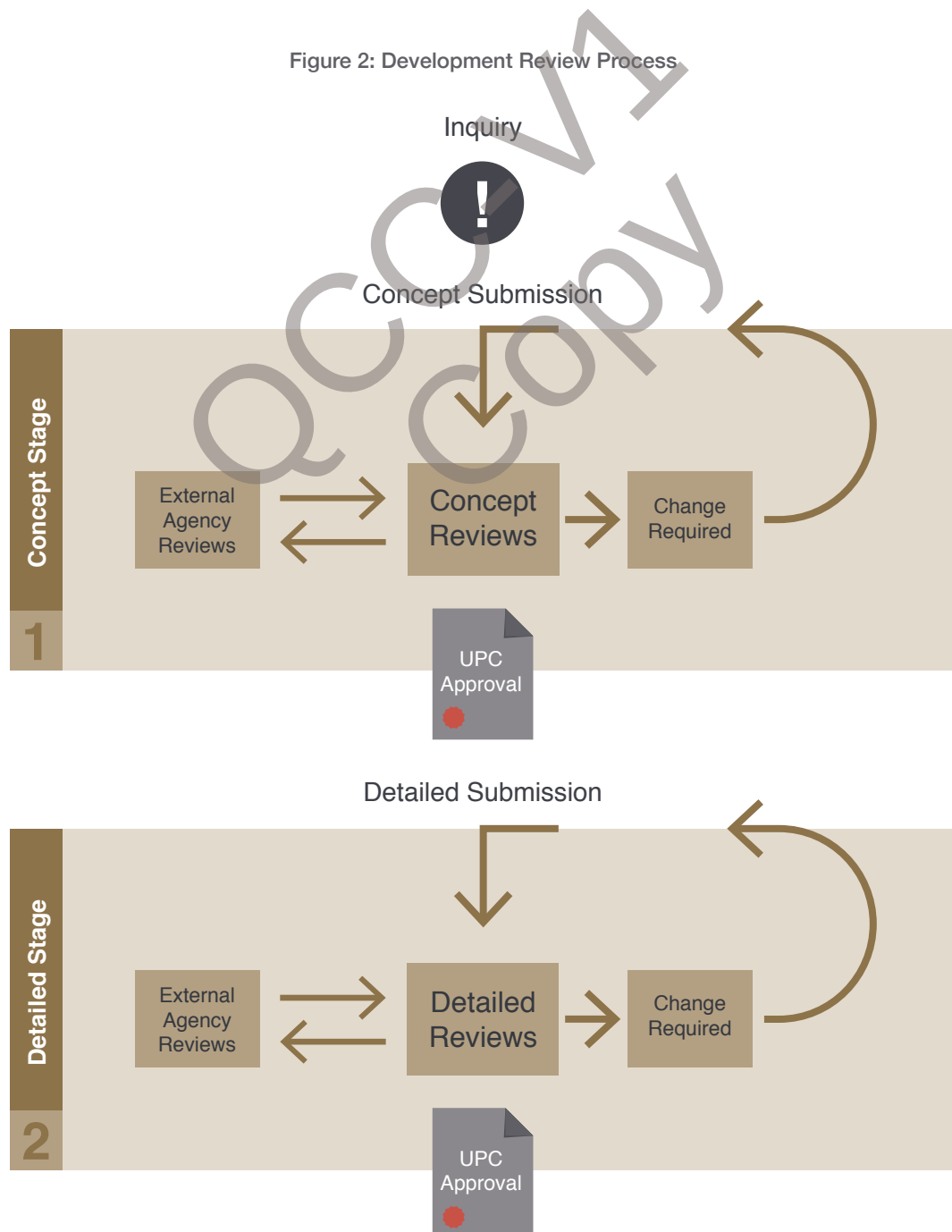
Figure 1: Development Review Streams

Master Plan	• Multiple Buildings • Road Network • Separate Multiple Buildings • Community Facilities Standards (applicable)
Project	• Single Building • No Road Network • Multiple Buildings With Connected Podium

As far as development planning is concerned, all developments, being either a project or master plan follow a two stage review process; these are Concept and Detailed stages. Each development is assessed for its compliance with existing plans and policies, best practice planning standards and other relevant government agency directives.

3.3 Development Review Process

The UPC development planning review process consists of two stages: Concept, and Detailed, as illustrated in Figure 2. Following initial enquiry by the developer, the approvals are given as part of the permitting system described in Section 3-3-1 through Section 3-3-3.



3.3.1 *Enquiry Meeting*

The development review process begins as soon as a development site is secured. The landowner or his/her representative attends a scheduled information meeting where UPC staff explain plans and policies that will determine site development potential and outline the development review and Estidama Integrated Design processes.

This enquiry meeting helps set the 'goal posts' (parameters), as guided by policy, for supportable development and alerts the applicant to documents/agencies that should be consulted.

Design Approach/Optional Workshops

The applicant uses UPC, other government agency input and site analysis (e.g. habitat, climate, available infrastructure) to prepare feasible design development options for the site. Consideration is given to general land uses and site layout (e.g. streets, block patterns, open space network and general massing). The applicant sets an Estidama vision for the project.

Development Review planners are committed to providing support to the applicant during this design process and remain available to assist the applicant through this journey.

3.3.2 *Step 1 - Concept Stage*

A concept application covers all of the systems-level components of a development concept. Concept applications include site and massing plans, a comprehensive approach to open space and community facilities, and proposals for sustainable infrastructure systems (e.g. water, energy, waste, storm water). The UPC and up to 20 other review agencies evaluate compliance with applicable plans and policies in order to agree on the seven key elements of any proposal:

- Land use;
- Density;
- Building form (e.g. height, massing, solar orientation, shading);
- Site layout/design (e.g. infrastructure networks, amount and location of public space);
- Services (e.g. utility, transportation and public community facilities);
- Strategies to achieve Estidama objectives and targets, and
- Phasing.

The project owner must also consider the requirement of a traffic impact study and traffic management plan in accordance with DMAT concept stage deliverables.

The concept approval provides a clear direction for the developer to progress to the detailed stage of the planning process, taking the necessary level of commitment to implement the vision of the development.

3.3.3 *Step 2 - Detailed Stage*

Projects

Applicants with projects prepare and submit detailed site and building designs for review.

- The UPC reviews;
- Exterior building form (e.g. massing, character);
- Site treatment (e.g. public realm);
- Circulation route;
- Landscaping;
- Shading and lighting;
- Signage; and
- Infrastructure (e.g. utility tie-in points, parking and loading).

The review focuses on context-sensitive design and high-quality public spaces and pedestrian amenities.

This step also confirms that any conditions of concept approval have been adequately met and incorporated accordingly.

Master Plan

Detailed Master Plans are intended to help applicants translate concept Master Plans into detailed regulations and guidelines that will control site-specific development over time. These detailed plans include the following:

- Plot-specific regulations for subdivision (e.g. plot sizes and dimensions);
- Land uses;
- Densities;
- Heights;
- Siting requirements (e.g. site coverage and setbacks);
- Parking and loading;
- Building massing and exterior form;
- Landscaping and public realm design; and
- Shading, lighting and signage.



Applicants also prepare design guidelines for building design, site design/ landscaping, public spaces, shading, lighting, universal access and signage.

Applicants also confirm a detailed phasing plan and development management plan that sets out construction, maintenance and service provision responsibilities.

3.4 Planning validation of Infrastructure and Road Alignment

For online validation of Road Alignment, Service Corridor Networks Routes and receiving “No Objection” for “NOI” (Notice of Intent) refer to e-services on the relevant municipality web page for online processing.

3.5 Planning Standards and Guidelines

The following documents set out the planning standards and guidelines applicable in the emirate of Abu Dhabi:

Sector –Specific Documents

Plan Capital 2030: Urban Structure Framework Plan	(DP-301)
Plan Al Ain 2030: Al Ain Urban Structure Framework	(DP-302)
Development Review Stream: Master Plan and Project	(DP-303)

Other References Documents

Community Facility Planning Standards	(DP-304)
Safety and Security Planning Manual	(DP-305)
Etidama Pearl Building Rating System: Design and Construction	(DP-306)
Etidama Pearl Community Rating System: Design and Construction	(DP-307)
Etidama Pearl Villa Rating System: Design and Construction	(DP-308)
Public Realm Design Manual (PRDM)	(PR-401)
Mosque Design Regulations	(PR- 405)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Urban Street Design Manual (USDM)	(ROW-603)
Utility Plot Design Manual (UPDM)	(ROW-604)

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PUBLIC REALM AND COMMUNITY **FACILITIES INFRASTRUCTURE**



4.1 Public Realm

4.1.1 Overview

The Public Realm includes all exterior spaces and linkages that are physically accessible and/ or visually accessible regardless of ownership. These elements include, but are not limited to the following:



Courtesy of UPC

Figure 3 public open space in Abu Dhabi (UPC)

- Streets;
- Pedestrian ways;
- Bikeways ;
- Bridges;
- Plazas;
- Nodes;
- Squares;
- Transportation hubs;
- Gateways;
- Parks;
- Waterfronts;

- Natural features;
- View corridors;
- Landmarks; and
- Building interfaces.

4.1.2 *Public Realm Categories*

To structure its planning and design, the public realm is separated into four categories: Parks, Streetscapes, Waterfronts and Public Places.

a) Parks:

Public spaces within a community for recreational use. Parks may include natural areas such as mountain ridges and wadi systems.

b) Streetscapes

The visual elements of a street including the road, sidewalk, street furniture, trees and open spaces that combine to form the street's character.

c) Waterfronts

All land areas along the water's edge.

d) Public Places:

All open spaces within a community visible to the public or for public gathering or assembly.

4.1.3 *Context in Abu Dhabi*

The Emirate of Abu Dhabi is planning for the next generation. The PRDM one piece of an Emirate-wide planning strategy. The PRDM builds on other planning documents in a continued effort to shape the future development of Abu Dhabi.



4.1.4 Objectives / Purpose of the Manual

The public realm in Abu Dhabi expresses traditional Arab culture while serving the diverse, multicultural population. It is a fully accessible and engaging experience that includes diverse public parks and civic spaces; an interconnected system of public walkways, bicycle trails and public transit; a vibrant and active waterfront; and active mixed-use areas that are all enhanced through high quality architecture, streetscape design and public art. It is safe, comfortable and responds to the climate and environment of Abu Dhabi.

The PRDM is focused on providing a public realm that meets the needs of the people of Abu Dhabi. Key to fulfilling this objective is understanding how the community views the public realm. During the planning process, various stakeholder meetings and a public survey were held to address community and agency needs. Further consideration has been given to the following in order to ensure the standards meet the specific requirements of the Emirate:

- **Quantity** - Targeting a total percentage of the defined urban areas to be set aside for parks, or protecting a total percentage of the land in any new development as open space;
- **Proximity** - Locating a park within a defined proximity of every resident;
- **Accessibility** - Ensuring parks are located to be physically accessible by foot, bicycle or public transit and visually accessible for the greater public;
- **Distribution** - Arranging park locations to ensure balanced services across geographical areas;
- **Equity** - Providing parks and open space evenly across diverse populations;
- **Coordination** - Combining park objectives with other development plans;
- **Balance** - Offering a mix of programming and activities throughout the park and open space system;
- **Shaping** - Identifying ways that the park and open space system can promote or contain growth; and
- **Connections** - Identifying ways to link parks and open spaces and associated resources.

Note - The PRDM quantifies Level Of Service only as it is applicable to the Park and Waterfront categories of the public realm.

4.1.5 Public Realm Principles

- The public realm has nine principles, including:
- Liveability;
- Identity;

- Access;
- Connectivity;
- Place making & Design Excellence;
- Environmental Stewardship;
- Inclusivity;
- Activation; and
- Shared Ownership & Implementation.

4.1.6 *Applicable Standards, Guidelines and Planning Documents*

The minimum quality and functionality of Public Realm is governed by the use of consistent and high quality standard treatments. The following documents are to be used for the planning, design and implementation of Public Realm to ensure its objectives are achieved:

Sector-Specific Documents

Public Realm Design Manual (PRDM)	(PR-401)
Lighting Manual	(PR-402)
Urban Street Design Manual (USDM)	(ROW-603)
Irrigation Manual (Volumes I) Design Manual	(WA-725-1)
Irrigation Manual (Volume II) Operation and Maintenance Manual	(WA-725-2)
Irrigation Manual (Volume III) Technical Specifications	(WA-725-3)
Irrigation Manual (Volume IV) Standard Drawings	(WA-725-4)

Other References Documents

Plan Capital 2030: Urban Structure Framework Plan	(DP-301)
Plan Al Ain 2030: Al Ain Urban Structure Framework	(DP-302)
Safety and Security Planning Manual	(DP-305)
Estidama Pearl Community Rating System: Design and Construction	(DP-307)
Abu Dhabi Surface Transport Master Plan	(TR-501)
Al Ain Surface Transport Master Plan	(TR-502)
Al Gharbia Surface Transport Master plan	(TR-503)
Eastern Region Surface Transport Master plan	(TR-504)



Addressing System Standards Manual (Volume II), Programme Design Policies, Authority and Organization	(TR-506-1)
Addressing System Standards Manual (Volume III) Geonames Standard and Practitioner's User Guide	(TR-506-2)
Addressing System Standards Manual (Volume IV), Street Addressing Standard and Practitioner's User Guide	(TR-506-3)
Addressing System Standards Manual (Volume V), Signage Standard and Practitioner's User Guide	(TR-506-4)
Manual on Uniform Traffic Control Devices	(TR- 511)
Walking and Cycling Master plan-Net Work Design	(TR-530)
Work Zone Traffic Management Manual	(TR-531)

4.2 Community Facilities

4.2.1 Overview

Community Facilities are facilities that provide a substantial benefit to the residents. They are fundamental to creating livable communities.

These Facilities include:

- Healthcare (e.g., clinics and hospitals);
- Education (e.g., nurseries, schools, colleges and universities);
- Governance and institutional services (e.g., Government administration, police and civil defence and post offices);
- Social, cultural and recreational facilities (e.g., community centres, libraries, cultural spaces and sports facilities);
- Religious facilities (e.g., Mosques). Mosque provision falls under the Abu Dhabi; and
- Mosque Development Regulations.

4.2.2 Context in Abu Dhabi

Community facilities (e.g., mosques, hospitals, schools and community centres) are important to achieve successful, healthy and sustainable communities served by a full range of community facilities that are appropriate to residents' needs and are accessible to all. However, the new planning standards developed by UPC require provision of facilities according to a new 6-level community facilities hierarchy. This new standard will lead to a growing development of community facilities to satisfy the needs of Abu Dhabi population.

4.2.3 Objectives

The Community Facilities Planning Standards (CFPS) were developed in collaboration with the Government agencies responsible for regulating and delivering community facilities. The minimum objectives of the Standards are to:

- Ensure that appropriate community facilities are designed and provided for all and that these facilities are appropriately sized and located;
- Ensure that all but the very smallest communities are served by a range of community facilities;
- Provide community facilities that make the best use of land, offer ease of access to users and allow flexibility for possible future demographic changes;
- Ensure that community facilities support Estidama objectives for the creation of sustainable communities; and
- Ensure that the need for community facilities is weighed equally against other spatial and land use requirements when planning sustainable communities across the Emirate.

4.2.4 Key Design Considerations

The provision of community facilities is crucial to providing a high quality of life. As such, community facilities design must be based on:

- Demographic assumptions to determine the size of the population that will require community facilities provision;
- Community Facility Standards tables which shall be applied to the resulting population calculations;
- Guidance on the appropriate level and scale of community facility provision depending on the location of the development proposal by type of settlement and type of centre; and
- Guidance on how facilities can be provided through co-located, integrated or adjacent facility designs which are responsive to the needs of the community.



4.2.5 *Applicable Standards, Guidelines and Planning Documents*

The documents in the following lists are to be used for the planning, design and construction of community facilities to ensure objectives, at a minimum, are achieved:

Sector-Specific Documents

Design Manual: Minimum Requirements for Private School Facilities	(PR- 403)
Design Manual: Standards and Criteria for School Facilities	(PR- 404)
Mosque Design Regulations	(PR-405)
Community Facility Planning Standards	(DP-304)
Safety and Security Planning Manual	(DP-305)
Estdama Pearl Community Rating System: Design and Construction	(DP-307)

Other References Documents

Plan Capital 2030: Urban Structure Framework Plan	(DP-301)
Plan Al Ain 2030: Al Ain Urban Structure Framework	(DP-302)
Estdama Pearl Building Rating System: Design and Construction	(DP-306)
Work Zone Traffic Management Manual	(TR-531)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)

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TRANSPORT **INFRASTRUCTURE**



5.1 Overview

The responsible Authority for transport in the Abu Dhabi Emirate is the Department of Municipal Affairs and Transport (DMAT). The transport infrastructure in the Emirate, covered in this document, includes the following components:



Figure 4 Shaikh Zayed Bridge

- Roads (highways and urban streets including bridges, viaducts, tunnels and underpasses);
- Land Public Transport (infrastructure and operation of services);
- Road traffic control and management;
- Pedestrian walkways, and Bicycle paths;
- Right of way of the above; and
- Parking Structures.

The following transport infrastructure components are not considered in this document, except where they interface with the above-mentioned components:

- Maritime Transport
- Aviation

Rail, Maritime, bus and aviation infrastructure standards are established and enforced by the DMAT. Any authority, organization, developer, consultant or contractor involved in, or affected by the planning, design, construction, maintenance or operation of this infrastructure shall liaise directly with the DMAT and other concerned Authorities.

Context in Abu Dhabi

Transport infrastructure is essential in supporting the economic growth of any nation, and provides connectivity and the efficient movement of people, goods and services. Abu Dhabi's road network has improved rapidly and progressively by providing highways, which serve as the links between cities, towns and outlying areas, for the use of residents and visitors alike. *

In 2012, some 2,526 km of external roads (outside of settlements) were available to connect different parts of the Emirate, which increased to 2,768 km at the end of 2015. *

Investment to maintain and expand the road network across Abu Dhabi increases in correlation to the number vehicles on the road. From 2005 to 2011, the number of licensed vehicles traveling on Abu Dhabi roads grew by 191 %. In 2011, the number of registered vehicles was 783,180, which increased to 947,508 in 2014, an increase of some 21%.

* DMAT: Department of Municipal Affairs and Transport.

5.2 Roads Infrastructure Standards

Highways include all roads, in the Emirate of Abu Dhabi, generally outside of urban areas, and freeways within the urban areas, while urban streets are within the existing and planned urban areas of the Emirate. Urban areas are defined in the maps contained in the Plan Capital 2030, Plan Al Ain 2030 and Plan Al Gharbia 2030 documents. All roads are under the jurisdiction of the DMAT.

5.2.1 Objectives

The road network of the Emirate provides accessibility and mobility for people, goods and services. The road hierarchy framework enables each road to provide a level of access and mobility, depending upon its position in the road hierarchy (strategic, arterial, distributor or local access).

The concept of the functional classification of roads is one of the most important determining factors in roadway design. The overall objective of the functional classification system, when viewed in its entirety, is to yield an optimum balance between its access and mobility purposes. Therefore, the functional classification system provides the framework for determining the design criteria for individual roads.

Within urban areas, roads and streets play a significant role in expressing the identity of the cities and neighbourhoods that they serve. In this sense urban streets are more than just conduits of movement; they help to create urban places and spaces for people to gather, stroll, and experience and enjoy city and neighbourhood life. Context is, therefore, a crucial factor that influences urban street design and in the particular the design of the spaces outside of the vehicular travel zones of urban streets.



5.2.2 Key Design Considerations and Principles

A range of factors influence design choices for road projects. The design characteristics and values that are adopted must provide satisfactory service to all users, and be economically viable within the financial, topographical and environmental constraints that may exist. With that in mind, some of the key design considerations are the following:

- To provide connectivity and accessibility;
- To design roads that create a safe environment for all users;
- To facilitate the transition from a society based on vehicle trips to a multi-modal transit community;
- To introduce a street network that allows for greater route and mode choices for pedestrians and cyclists and improves the capacity and efficiency of the Emirate of Abu Dhabi's urban streets;
- To be appropriate for the land use context;
- To provide public transport (for example, bus and rail) services corridors within their RoW;
- To provide public transport priority measures within the RoW;
- To provide utility service corridors within their Row; and
- To avoid the terrestrial and marine protected areas and avoid/reduce overall environmental impacts throughout the project or masterplan life cycle (Planning, Design and Construction and operation).

The design of road networks is based on the following (fundamental) principles:

- The best transport plan is inter-related to a good land-use plan;
- Good road design will take into consideration public transit along with pedestrians and cyclists;
- A well designed road network provides safety for all modes of transport;
- Roads connectivity enhances capacity and allows smooth traffic flow;
- Road design shall reflect Plan 2030 and emerging Plan Capital 2030 goals for the Abu Dhabi Emirate;
- Road design will take into account the Surface Transport Masterplans for Abu Dhabi Region, Eastern Region and Al Gharbia; and
- Road design shall consider Abu Dhabi Emirate sustainability principles.

5.2.3 *Applicable Standards, Guidelines and Planning Documents*

All roads, irrespective of their classification shall be planned, designed, constructed, operated and maintained and in accordance with the following standards and guidelines:

Sector-Specific Documents

Abu Dhabi Surface Transport Master Plan	(TR-501)
Al Ain Surface Transport Master Plan	(TR-502)
Al Gharbia Surface Transport Master plan	(TR-503)
Eastern Region Surface Transport Master plan	(TR-504)
Freight Master Plan - Truck Parking Standards	(TR-505)
Addressing System Standards Manual (Volume II), Programme Design Policies, Authority and Organization	(TR-506-1)
Addressing System Standards Manual (Volume III) Geonames Standard and Practitioner's User Guide	(TR-506-2)
Addressing System Standards Manual (Volume IV), Street Addressing Standard and Practitioner's User Guide	(TR-506-3)
Addressing System Standards Manual (Volume V), Signage Standard and Practitioner's User Guide	(TR-506-4)
Congestion Management Policy and Procedures	(TR-507)
Environmental Impact Assessment Guidelines for Road Projects	(TR-508)
Geotechnical Investigation and Design (Volume I) Managing Geotechnical Risk	(TR-509-1)
Geotechnical Investigation and Design (Volume II) Ground Investigation and Geotechnical Design	(TR-509-2)
Land Surveying and Mapping Guide for Road Projects	(TR-510)
Manual on Uniform Traffic Control Devices	(TR-511)
Non-Disruptive Road Crossings for utilities Manual	(TR-512)
Pavement Design Manual	(TR-513)
Road Geometric Design Manual	(TR-514)
Road Performance Monitoring Systems Guide	(TR-515)
Road Structures Design Manual	(TR-516)
Road user code	(TR-517)
Roadside Design Guide	(TR-518)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Technical circular –Bus Stop Design standard drawings	(TR-520)
Guide Drawings on locating Bus stops on Urban Roads	(TR-521)



Technical Guidance Document for Construction Environmental Management Plan	(TR-522)
Technical Guidance Document for Environmental Audit Report	(TR-523)
Technical Guidance Document for Environmental Impact Assessment	(TR-524)
Technical Guidance Document for Preliminary Environmental Review	(TR-525)
Technical Guidance Document for Strategic Environmental Impact Assessment	(TR-526)
Technical Guidance Document for Submission of Environmental Permit Applications and Reports	(TR-527)
Traffic Calming Guideline	(TR-528)
Traffic Signals and Electronic Warning Information Systems Manual	(TR-529)
Walking and Cycling Master Plan – Network Design	(TR-530)
Work Zone Traffic Management Manual	(TR-531)
Access Management Policy & Procedures	(TR-532)
Plan Capital 2030: Urban Structure Framework Plan	(DP-301)
Plan Al Ain 2030: Al Ain Urban Structure Framework	(DP-302)
Lighting Manual	(PR-402)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for Utilities in urban areas)	(ROW-602)
Urban Street Design Manual (USDm)	(ROW-603)
Irrigation Manual (Volume I) Design Manual	(WA-725-1)
Irrigation Manual (Volume II) Operation and Maintenance Manual	(WA-725-2)
Irrigation Manual (Volume III) Technical Specifications	(WA-725-3)
Irrigation Manual (Volume IV) Standard Drawings	(WA-725-4)
Stormwater and subsoil drainage systems (Volume I) Design Manual	(WA-726-1)
Stormwater and subsoil drainage systems (Volume II) Standard Drawings	(WA-726-2)
Stormwater and subsoil drainage systems (Volume III)Operation and Maintenance Management Manual	(WA-726-3)
Stormwater and subsoil drainage systems Standard Technical Specification (Volume 1, 2 & 3)	(WA-726-4)

Other References Documents

Safety and Security Planning Manual	(DP-305)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)

5.3 Land Public Transport Infrastructure Standards

Public transport infrastructure is defined as the infrastructure developed to carry the public in modes of shared transport provided for community use operating a scheduled timetable and regulated fare structure. This includes the following modes:

- Taxi;
- Bus;
- Bus Rapid Transit (BRT)
- Light Rail / Tram;
- Metro; and
- Regional Rail.

For the purpose of this document, marine and aviation infrastructure will not be discussed here. Marine and aviation infrastructure standards are established and enforced by DMAT.

5.3.1 Objectives

The purpose of public transport infrastructure is to provide the community with a sustainable, efficient, convenient and safe transportation alternative to private vehicles. A public transport system must enhance the community it serves and be easily accessible by commuters, on foot or by bike and provide easy way-finding for customers throughout the entire duration of their journey. For a public transport system to be successful, it must provide a desirable alternative to private vehicle use.

5.3.2 Key Design Considerations

Effective public transport infrastructure shall be developed by considering the following principles in the design and construction of such facilities:

- Connectivity and Accessibility;
- Mobility;
- Convenience;
- Efficiency;
- Safety for all users;
- Visibility;
- Aesthetics;
- Sustainability; and
- Integration with other modes of transport



5.3.3 *Applicable Standards, Guidelines and planning documents*

All public transport infrastructures shall be planned and safeguarded in accordance with the following sector-specific requirements, standards and guidelines:

Bus Rapid Transit Planning Standards	(TR-533)
Bus Services Planning Standards	(TR-534)
Metro Planning Standards	(TR-535)
Railway Planning Standards	(TR-536)
Tramways Planning Standards	(TR-537)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Technical circular –Bus Stop Design standard drawings	(TR-520)
Guide Drawings on locating Bus stops on Urban Roads	(TR-521)
Technical Guidance Document for Submission of Environmental Permit Applications and Reports	(TR-527)
Walking and Cycling Master Plan -Network Design	(TR-530)

The Design, Construction, Operation and Maintenance of the emerging public transport infrastructures shall be in accordance with the specific requirements and Standards developed during the different stages of any such project.

The design and construction of bus infrastructure (bus stations, bus depots, bus shelters etc.) will be in accordance with the specific requirements and standards for implementation of the DMAT plans for the development of the bus network and infrastructure in the emirate.

5.4 Road Traffic Control and Management Infrastructure Standards

5.4.1 *Objectives*

The aim of Road Traffic Control and Management Infrastructure is to maximise the efficiency, safety and convenience of the road network. These goals are usually achieved by developing integrated transport networks utilising international best practice and a consistent set of standards and methodologies. Consistency of the traffic control and management throughout the transport networks is vital for ensuring that users have a clear understanding of the network, and can plan their journey accordingly.

5.4.2 Key Design Considerations

There are many reasons to implement traffic control and management, for instance; congestion relief, safety, work zone and events management etc. While these reasons may change, the (fundamental) design considerations remain the same and can be summarized as below:

- Efficiency;
- Safety;
- User needs;
- Integration with other modes of transport; and
- Data collection and distribution to users.

5.4.3 Applicable Standards, Guidelines and Planning Documents

All road traffic control and management infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following requirements, standards and guidelines:

Sector-Specific Documents

Abu Dhabi Surface Transport Master Plan	(TR-501)
Al Ain Surface Transport Master Plan	(TR-502)
Al Gharbia Surface Transport Master plan	(TR-503)
Eastern Region Surface Transport Masterplan	(TR-504)
Manual on Uniform Traffic Control Devices	(TR-511)
Road Geometric Design Manual	(TR-514)
Road Performance Monitoring System Guide	(TR-515)
Road Structures Design Manual	(TR-516)
Roadside Design Guide	(TR-518)
Traffic Signals and Electronic Warning and Information Systems Manual	(TR-529)
Work Zone Traffic Management Manual	(TR-531)
Urban Street Design Manual (USDMM)	(ROW-603)

Other References Documents

Safety and Security Planning Manual	(DP-305)
Public Realm Design Manual (PRDM)	(PR-401)



5.5 Pedestrian Walkways, And Bicycle Paths

5.5.1 Objectives

Liveable communities are often categorised as having well-integrated, safe, and inviting pedestrian and cyclist spaces as part of their infrastructure. These facilities provide the following functions:

- Commuting and utility trips for children and adults;
- Transport over short distances, for instance to schools;
- Access to other transport modes including Public Transit for longer journeys;
- Access to buildings and facilities;
- Recreation;
- Encouragement of an active and healthy lifestyle;
- Encouragement of community integration; and
- Safe road and street crossings.

The overall objective of creating liveable communities can be achieved when these functions of pedestrian and bicycle infrastructure are consistently integrated into the Public Realm environment, connecting key land uses and interfaced with Public Transit.

5.5.2 Key Design Considerations

Pedestrians and cyclists are the most vulnerable road users. Pedestrians are also the most important, as all road users are pedestrians at some point in their journey. As such, pedestrian and cyclist-specific infrastructure should be designed to the highest standard possible with the following key design considerations in mind:

- Safety;
- User comfort and security, for example lighting and secure cycle parking;
- Connectivity and Accessibility;
- Mobility (including the mobility of (physically challenged persons);
- Shading and cooling;
- Way finding;
- Aesthetics;
- Pedestrians and cyclists volume;
- Appropriate walking and cycling distances; and
- Crime Prevention through Environmental Design (CPTED) principles.

5.5.3 *Applicable Standards, Guidelines and planning documents*

All pedestrian walkways and bicycle paths shall be planned, designed, constructed, operated and maintained in accordance with the following requirements, standards and guidelines:

Sector-Specific Documents

Abu Dhabi Surface Transport Master Plan	(TR-501)
Al Ain Surface Transport Master Plan	(TR-502)
Al Gharbia Surface Transport Master plan	(TR-503)
Eastern Region Surface Transport Masterplan	(TR-504)
Manual on Uniform Traffic Control Devices	(TR-511)
Road Geometric Design Manual	(TR-514)
Road Structures Design Manual	(TR-516)
Traffic Signals and Electronic Warning and Information Systems Manual	(TR-529)
Walking and Cycling Master Plan -Network Design	(TR-530)
Public Realm Design Manual (PRDM)	(PR-401)
Urban Street Design Manual (USDm)	(ROW-603)

Other References Documents

Plan Capital 2030: Urban Structure Framework Plan	(DP-301)
Plan Al Ain 2030: Al Ain Urban Structure Framework	(DP-302)
Safety & Security Planning Manual	(DP-305)





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6

RIGHT OF WAY AND **UTILITY DISPOSITION**



6.1 Overview

Right of Way means the area of land appropriated for providing existing and planned land transportation infrastructure and accommodating utility services, bridges or tunnels, and other public realm components over which or through which land transportation infrastructure passes, and shall include the appurtenant height above and depth below the land and equipment installed on such land. In the event that the area of land is bordered by plot boundaries, it shall include the land between such plot boundaries.



Courtesy of UPC

Figure 5: under construction street according to UCDM

Utility Corridor, as a part of Right of Way is an area of land or a volume of space that is allocated with the intention of accommodating services utilities.

6.2 Context in Abu Dhabi

The Abu Dhabi Emirate has developed a series of guidelines to allocate the utilities within the Utilities Corridor of the Right of Way for the purpose of providing services utilities for all areas of development. In Abu Dhabi, like other countries throughout the world, the Utility Corridor commonly runs alongside roadways, but even goes beyond that to occupy more than 20,000 km of linear space. This strategic location helps to provide accessibility for installation, but also operation and maintenance of the utilities that are vital for sustained, yet productive, growth.*

* UPC: Urban Planning Council.

6.3 Objectives

The objective of the Utility Corridor is to provide well-planned and appropriately sized provisions for utilities along the road corridors or other areas to provide services to public. A set of planning and design standards have been developed to guide the design of these corridors to ensure that all utilities are represented and given adequate, but not excessive, space.

6.4 Key Design Considerations

In order to achieve the above objectives, key consideration must be integrated into the design:

- Functionality to serve the purpose;
- Appropriate hierarchy of utilities;
- Accessibility for construction and/or maintenance; and
- Protection of utilities.

6.5 Applicable Standards and Guidelines

All Utility Corridors shall be planned in accordance with the following references to ensure the fulfilment of the above-mentioned objectives and to serve the intended communities and the environment.

Sector-Specific Documents

Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)

Other References Documents

Urban Street Design Manual (USDM)	(ROW-603)
Utility Plot Design Manual (UPDM)	(ROW-604)
Safety and Security Planning Manual	(DP-305)
Public Realm Design Manual (PRDM)	(PR-401)
Road Geometric Design Manual	(TR-514)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)





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7

INTEGRATED WATER MANAGEMENT **INFRASTRUCTURE**



7.1 Overview

This section provides a brief overview of the water management infrastructure sector in the Emirate of Abu Dhabi. For this purpose, the water management infrastructure sector is divided into the following sub-sectors, namely:

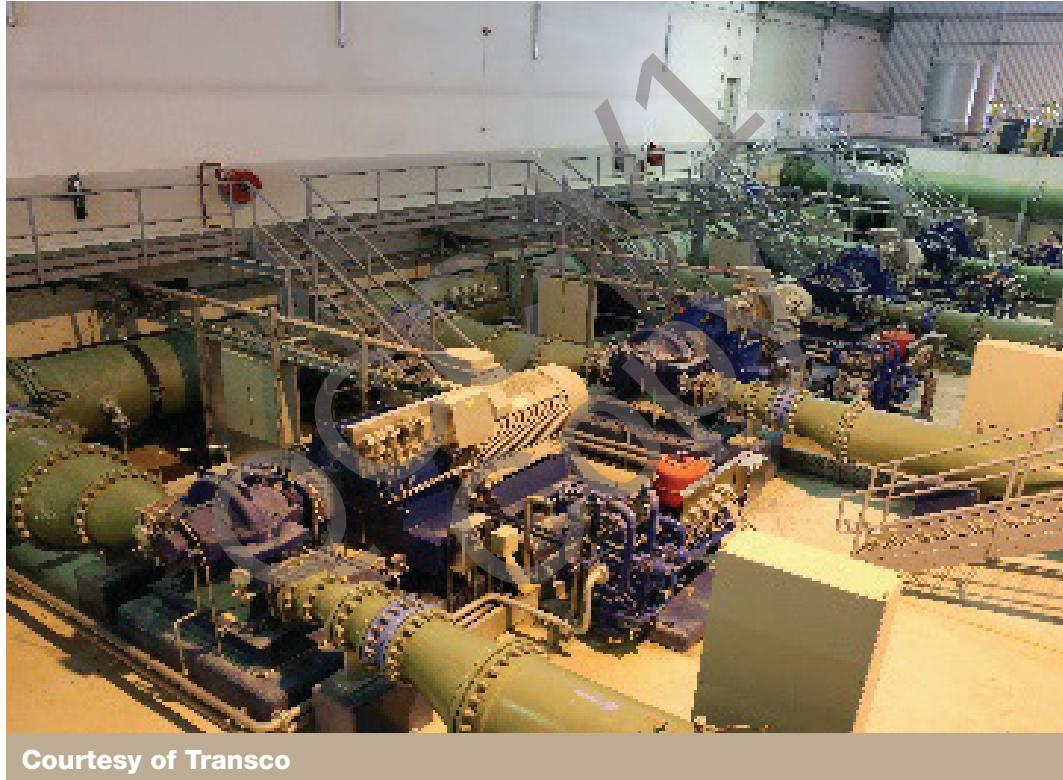


Figure 6: Potable Water Pumping Station

- Potable water transmission infrastructure;
- Potable water distribution infrastructure;
- Waste water infrastructure;
- Irrigation water infrastructure;
- Storm water infrastructure; and
- Ground water monitoring infrastructure.

7.2 Context in Abu Dhabi

Water production in Abu Dhabi is mainly based on desalination. In 2014 thermal desalination accounted for 92.4% of water production in Abu Dhabi, predominantly via multi-stage flash (MSF) distillation, and to a lesser extent multiple effect distillation (MED), while the remaining 7.6% was generated by seawater reverse osmosis (RO). Total gross production capacity was 916 MIGD (4.16 million m³ per day). The 3,326km water transmission system carried an average of 760 MIGD

(3.5 million m³ per day) of desalinated water via mains pipelines. The drinking water transmission network comprises 50 pumping stations with a combined capacity of 2,436 MIGD, and 122 reservoirs with a total capacity of 655 MIG. In 2014, the average daily water supply was 184.5 MIGD to Al Ain and 495 MIGD to Abu Dhabi and the total length of the distribution system operated by the Al Ain and Abu Dhabi distribution companies (e.g., AADC and ADDC) was about 12,253 km. Their water network comprises a total of 47 pumping stations with an overall capacity of 122 MIGD, and 110 reservoirs with a total capacity of 96 MIG. By the end of 2014, there were 349,492 water customers (i.e., connections) in the Emirate of Abu Dhabi.*

The second source of water in Abu Dhabi is groundwater; it is contained in consolidated or unconsolidated surficial deposit aquifers and bedrock/structural aquifers. The expansion of agriculture and forestry in the Emirate has been recognized as primary factors responsible for depletion of this finite source. Together, these two sources have consumed about 70% of Abu Dhabi's ground water resources.*

Repurposing wastewater following treatment has been identified as one of the strategies for water source diversification that can help mitigate Abu Dhabi's water shortage. Abu Dhabi has 39 wastewater treatment plants that process approximately 944,000 m³ of wastewater that is generated every day. In 2015, a total of 344 million m³ was treated; from this, 51% was reused mainly for irrigation of green areas.*

* RSB: Regulation and Supervision Bureau.

7.3 Potable Water Transmission Infrastructure Standards

Potable water transmission infrastructure is operated and maintained by TRANSCO, a subsidiary of ADWEA. The standards, manuals and guidelines associated with the planning, design, construction and operation of potable water transmission infrastructure are listed in this chapter.

7.3.1 Objectives

The objectives of potable water transmission infrastructure, operated by TRANSCO, are to reliably, securely and safely transmit water from the water producers to the distribution networks operated by ADDC, AADC and to the Northern Emirates of the UAE, maintaining the optimum balance of performance, risk, cost and sustainable development.

To execute these objectives, TRANSCO maintains and operates a network of transmission assets, including Pumping Stations, Reservoirs, Water trunk mains, Water Control Centre and undertakes the management of all planned and unplanned maintenance activities in accordance with defined policies & procedures.



7.3.2 Key Design Considerations

The supply of potable water to a community is a necessity of life, and as such the transmission infrastructure must be reliable, secure and safe. In order to achieve this objective, a strict set of standards are applied and key design consideration are as follows:

- Adequate water pumping and storage facilities;
- Wholesome potable water supply;
- Safety of community;
- Security;
- Reliability;
- Durability;
- Sustainability;
- Accessibility for maintenance;
- Safety of maintenance teams;
- Efficiency of planned and unplanned maintenance;
- Demand;
- Supply; and
- Minimise leakage and wastage.

7.3.3 Applicable Standards and Guidelines

Potable water transmission infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following requirements, standards and guidelines:

Sector-Specific Documents

Drinking Water Safety Plan	(WA-701)
Hygienic Practices within Pumping Station and Water Network	(WA-702)
Metering and Data Exchange Code	(WA-703)
Standard Technical Specifications for Water Works	(WA-704)
The Water Quality Regulations	(WA-705)
The Water Transmission Code	(WA-706)

Other References Documents

Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)

7.4 Potable Water Distribution Infrastructure Standards

Potable water distribution infrastructure is operated and maintained by ADDC and AADC in their respective jurisdictions; both are subsidiaries of the ADWEA. The standards, manuals and guidelines associated with the planning, design, construction, operation and maintenance of potable water distribution infrastructure are listed in this chapter.

7.4.1 Objectives

The objectives of potable water distribution infrastructure, operated by ADDC in Abu Dhabi and Western Region and AADC in Al Ain, is to provide safe, efficient, and economical potable water distribution to the community. As such, ADDC and AADC design, develop, operate and maintain a system of water distribution infrastructure according to a strict set of standards.

7.4.2 Key Design Considerations

The objectives from Section 7.4.1 can be achieved through the application of these strict standards and key design consideration:

- Adequate water pumping and storage facilities;
- Wholesome potable water supply;
- Safety of community;
- Security;



- Reliability;
- Durability;
- Sustainability;
- Accessibility for maintenance;
- Safety of maintenance teams;
- Efficiency of planned and unplanned maintenance;
- Demand / supply; and
- Minimise leakage and wastage.

7.4.3 *Applicable Standards and Guidelines*

All potable water distribution infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following requirements, standards and guidelines:

Sector-Specific Documents

The Water Supply Regulations	(WA-708)
Guide to Water Supply Regulations	(WA-709)
The Water Distribution Code	(WA-710)
The Customer Metering Regulations	(WA-711)
Metering and Data Exchange Code	(WA-703)
Standard Technical Specifications for Water Works	(WA-704)
The Water Quality Regulations	(WA-705)
The water Transmission Code	(WA-706)

Other References Documents

Guideline for Work in Vicinity of ADDC Assets	(WA-712)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual.	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)

7.5 Waste Water Infrastructure Standards

ADSSC is responsible for establishing and applying all standards and specifications related to waste water infrastructure in the Emirate of Abu Dhabi.

7.5.1 Objectives

The objectives of waste water infrastructure is to collect and treat waste water discharged from all residential and commercial customers in the Emirate of Abu Dhabi and safely dispose of both the recycled water and bio-solids produced thereof. This includes the following:

- Waste water collection;
- Waste water treatment; and
- Recycled water & bio-solids disposal.

7.5.2 Key Design Considerations

Key design considerations must be applied during the collection and treatment of wastewater to produce recycled water and bio-solids. Those considerations are as follows:

- Environmental responsibility;
- Sustainability;
- Safety;
- Efficiency of maintenance and operation; and
- Demand.

7.5.3 Applicable Standards and Guidelines

All waste water infrastructure shall be planned, designed, constructed, operated and maintained in according to the following sector-specific requirements, standards and guidelines:



Sector-Specific Documents

Design Guidelines	(WA-713)
General Specification for Civil Works	(WA-714)
General Specification for Electrical Works	(WA-715)
General Specification for Mechanical Works	(WA-716)
Standard and Typical Drawings	(WA-717)
Standard Method of Measurement	(WA-718)
Technical Data Sheet for Electrical Works	(WA-719)
Technical Data Sheet for Mechanical Works	(WA-720)
The Trade Effluent Control Regulations	(WA-721)
Recycled Water and Bio-solids Regulations	(WA-722)
Guide to the Recycled Water and Bio-solids Regulations	(WA-723)
Guide to the Trade Effluent Control Regulations	(WA-724)

Other References Documents

Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)

7.6 Irrigation Water Infrastructure

Recycled water can be used for irrigation purposes within the Emirate of Abu Dhabi. Recycled water, also known as TSE, is a by-product of the ADSSC's waste water treatment process, and can be used for a number of applications. It is, however, currently only used for irrigation and landscaping purposes.

ADSSC has jurisdiction over irrigation water/recycled water main transmission infrastructure. These main transmission lines feed into storage facilities or tie into distribution networks, which are both overseen by DMAT.

If the irrigation infrastructure is designed and constructed by a developer, it might be handed over to DMAT once it is commissioned, subject to compliance to DMAT requirements for design and construction.

7.6.1 *Objectives*

There is also a push for authorities to provide more sustainable infrastructure, and operate in a more sustainable manner. With this in mind the objectives of irrigation water infrastructure are to provide a well maintained and sustainable irrigation system that utilises the latest water conserving technologies to irrigate sustainably landscaped areas within the Public Realm and road reserve.

7.6.2 *Key Design Considerations*

Irrigation water demand is greatly influenced by the landscaping design; therefore, it is imperative that the irrigation infrastructure is designed in conjunction with the landscaping. Key design considerations for the design of irrigation water infrastructure are as follows:

- Recycled water quality;
- Safety of community;
- Security;
- Reliability;
- Accessibility for maintenance;
- Safety of maintenance teams;
- Demand / supply;
- Sustainability;
- Environmental Requirements;
- Landscape design (low water use plants); and
- Water conserving irrigation technologies.



7.6.3 *Applicable Standards and Guidelines*

Irrigation water infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following requirements, standards and guidelines:

Sector-Specific Documents

Irrigation Manual (Volume I) Design Manual	(WA-725-1)
Irrigation Manual (Volume II) Operation and Maintenance Manual	(WA-725-2)
Irrigation Manual (Volume III) Technical Specifications	(WA-725-3)
Irrigation Manual (Volume IV) Standard Drawings	(WA-725-4)

Other References Documents

Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Recycled Water and Bio-solids Regulations	(WA-722)
Guide to the Recycled Water and Biosolids Regulations	(WA-723)
Public Realm Design Manual (PRDM)	(PR-401)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Urban Street Design Manual (USDM)	(ROW-603)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)

7.7 Stormwater Infrastructure

Stormwater infrastructure is owned and operated by DMAT. If the infrastructure is designed and constructed by a developer, it might be handed over to DMAT once it is commissioned, subject to compliance to DMAT requirements for design and construction.

7.7.1 Objectives

The objectives of stormwater infrastructure are to collect, and transfer stormwater runoff in an efficient and sustainable manner. Recently a new approach of stormwater ponds with recreational facilities has been introduced.

7.7.2 Key Design Considerations

In order to improve the impact of the built environment on the natural environment, stormwater design should be developed with sustainability in mind. In principle, a sustainable stormwater design attempts to mimic the natural conditions as closely as practical. To develop a sustainable stormwater system the following key design considerations should be integrated:

- Water volume (runoff or surface water);
- Water velocity;
- Infiltration rates;
- Maintenance and sand control;
- Sustainability;
- Environmental Requirements;
- Treatment of contaminants;
- Safety; and
- Economics.

7.7.3 Applicable Standards and Guidelines

Stormwater infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following requirements, standards and guidelines:

Sector-Specific Documents

Stormwater and subsoil drainage systems (Volume I) Design Manual	(WA-726-1)
Stormwater and subsoil drainage systems (Volume II) Standard Drawings	(WA-726-2)
Stormwater and subsoil drainage systems (Volume III) Operation and Maintenance Management Manual	(WA-726-3)
Stormwater and subsoil drainage systems Standard Technical Specification (Volume 1, 2 & 3)	(WA-726-4)



Other References Documents

Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Urban Street Design Manual (USDM)	(ROW-603)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)

7.8 Ground Water Monitoring Infrastructure

Groundwater is the water held in the pores and fractures of underground soil and rocks, and known as the Aquifer. Environment Agency – Abu Dhabi (EAD) is responsible for groundwater research in the Emirate of Abu Dhabi, including exploration and monitoring. The EAD's jurisdiction also covers protection of groundwater through establishing and enforcing relevant policies and legislation to retain this precious and strategic resource.

Groundwater can be used as source of irrigation water in the farms, forests, and some industrial and constructional projects.

7.8.1 Objectives of Ground Water Infrastructure

The main objectives of groundwater infrastructure are to assess and protect groundwater, which can be done through exploration and monitoring activities. The exploration activities aim to assess all aquifer system, in terms of quantity and quality, by drilling exploration wells. Conversely, monitoring activities aim to observe any changes in groundwater quantity and quality caused by abstraction activities and land uses, which is carried out via the drilling of monitoring wells.

7.8.2 Key Design Considerations

The overall objectives of the design are focused on exploring and protecting the groundwater, considering the following activities:

- Conduct comprehensive exploration program in terms of geographic distribution and vertical aquifer systems;

- Conduct comprehensive network monitoring for various geographic and aquifer systems;
- Implement an effective design to protect the groundwater infrastructure wells, ensuring integrity of the network, as well as continuity of the operations and data flow;
- Implement an effective action plan for operating and maintaining the groundwater stations; and
- Utilize latest technologies in the groundwater operations.

7.8.3 *Applicable Standards and Guidelines*

Groundwater monitoring infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following:

Standard Operating Procedures for Permitting of Development and Infrastructure Projects.

(TR-519)





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ENERGY INFRASTRUCTURE



8.1 Overview

This section provides a brief overview of the energy infrastructure sector in the Emirate of Abu Dhabi. This sector includes:

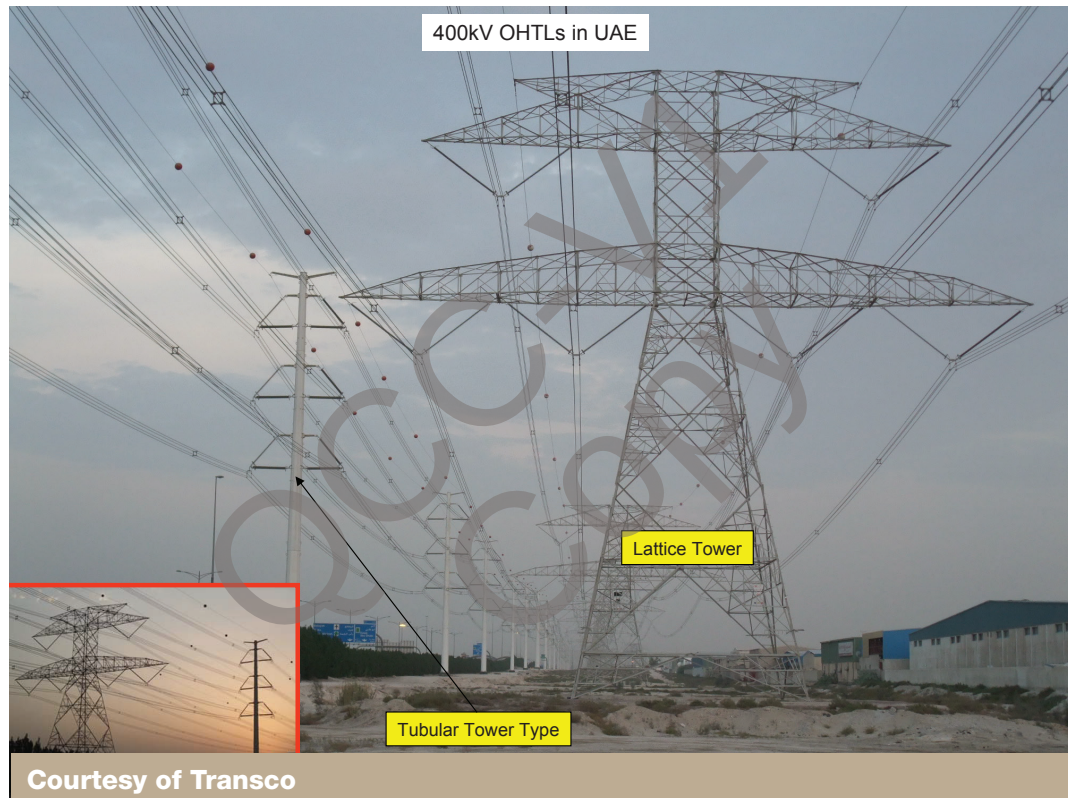


Figure 7: Electrical Transmission Lines

- Electricity Transmission Infrastructure;
- Electricity Distribution Infrastructure;
- Sustainable Energy;
- Road Lighting Infrastructure; and
- Public Lighting Infrastructure;

8.2 Context in Abu Dhabi

There are three separate services in the electricity sector: production, transmission and distribution. In 2014; total installed production capacity reached '15,5' Gigawatt (GW) which represents an increase of 79% since 2007. And the total electricity production was 70,847 Gigawatt hours (GWh), which represents an increase of 84% since 2007. Electrical productions correlates similarly with consumption, which also experienced a steep increase of 85% between 2007 and 2014 reaching a total consumption of 52,841 GWh in 2014. The difference

in production and consumption is common in this industry, and can likely be attributed to the power exported and line loss.*

* TRANSCO: Abu Dhabi Transmission and Depatch Company.

8.3 Electricity Transmission Infrastructure Standards

Electricity transmission infrastructure is operated and maintained by TRANSCO, a subsidiary of ADWEA. The standards, manuals and guidelines associated with the planning, design, construction, operation and maintenance of electricity transmission infrastructure are listed in this chapter.

8.3.1 Objectives

The purpose of electricity transmission infrastructure, operated by TRANSCO, is to reliably, securely and safely transmit electricity from the electricity producers, also known as Independent Water and Power Producers (IWPPs) to the distribution networks operated by ADDC and AADC, Northern Emirates Power Network and other Non-Embedded customers.

For this purpose, TRANSCO maintains and operates a network of transmission assets, including OHLs, Power Substations and the control centre 'LDC', and undertakes the management of all planned and unplanned, preventive and predictive maintenance activities in accordance with a set of clear standards.

8.3.2 Key Design Considerations

The supply of electricity to communities is necessity to maintain the quality of life in Abu Dhabi; therefore, the transmission infrastructure must be reliable, secure and safe. In order to achieve these requirements the following criteria are considered:

- Safety of community;
- Security;
- Reliability;
- Accessibility for maintenance;
- Safety of maintenance teams;
- Efficiency of planned and unplanned maintenance; and
- Demand / supply.



8.3.3 *Applicable Standards and Guidelines*

Electricity transmission infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following standards and guidelines:

Sector-Specific Documents

The Electricity Transmission Code	(EN-801)
Electricity Transmission Security Standard	(EN-802)
Standard Technical Specifications for Electricity 400-220-132-33-22-11 kV and Low Voltage Auxiliary System	(EN-803)
Electricity Design Guidelines of ADDC (5) year Planning Statement	(EN-804)
Metering and Data Exchange Code	(WA-703)

Other References Documents

Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)

8.4 Electricity Distribution Infrastructure Standards

Electricity distribution infrastructure is operated and maintained by ADDC and AADC in their respective jurisdictions; both are subsidiaries of the ADWEA. The standards, manuals and guidelines associated with the planning, design, construction, operation and maintenance of electricity distribution infrastructure are listed in this chapter.

8.4.1 Objectives

The purpose of electricity distribution infrastructure, operated by ADDC and AADC, is to provide safe, efficient and economical electricity distribution to the community.

8.4.2 Key Design Considerations

The supply of electricity to communities is necessary to maintain the quality of life in Abu Dhabi; therefore, the distribution infrastructure must be reliable, secure and safe. In order to achieve these requirements the following criteria are considered:

- Safety of community;
- Security;
- Reliability;
- Accessibility for maintenance;
- Safety of maintenance teams;
- Efficiency of planned and unplanned maintenance; and
- Demand / supply.



8.4.3 *Applicable Standards and Guidelines*

Electricity distribution infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following sector-specific standards and guidelines:

Sector-Specific Documents

Electricity Distribution code	(EN-806)
Electricity Wiring Regulation Code	(EN-807)
Engineering Standards for Earthing	(EN-808)
LV Design Guidelines	(EN-809)
Standard Technical Specification for Electrical Distribution Assets for 33KV, 22KV, 11 KV and Low Voltage Systems	(EN-810)
Electricity Design Guidelines of ADDC (5) year Planning Statement	(EN-804)
Metering and Data Exchange Code	(WA-703)

Other References Documents

Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)

8.5 Sustainable Energy

Sustainable energy is energy produced by means of renewable forms (wind, solar, hydro, etc.). Masdar is a developer and operator of renewable power generation projects within Abu Dhabi, and therefore determines the international standards appropriate for use in the UAE. Masdar makes direct investments in individual projects in all areas of renewable energy, with a focus on concentrating solar power, photovoltaic solar energy, as well as on- and offshore wind energy.*

* MASDAR.

8.5.1 Objectives

The objective of sustainable energy is to develop a means of power generation that utilizes renewable sources and does not diminish the environment for future generations. It is important for sustainable energy infrastructure to be developed to supplement current power generation methods, and develop into primary power generation means in the long term.

8.5.2 Key Design Considerations

Due to the size and complexities associated with the design, construction and maintenance of sustainable energy systems the following is a list of key design considerations for this infrastructure:

- Meet the needs of the community it services;
- Land and location restrictions;
- Climatic considerations in system selection
- Community acceptance of large infrastructure;
- Environmental impact assessments & Construction Environmental management;
- Projected GHG emissions displacement against 'traditional' power generation methods; and
- Ease of operations and maintenance.

8.5.3 Applicable Standards and Guidelines

Design of sustainable energy systems, while well-documented and developed, is relatively new to the UAE; therefore, no local standards and guidelines associated with it currently exist. Foreign and international best practice standards and guidelines are used in the UAE. Any authority, developer, consultant, contractor or other stakeholder that is engaged in any project where sustainable energy sources will be utilized can approach ADWEA and Masdar for advice and guidance.



8.6 Road Lighting Infrastructure Standards

Road lighting infrastructure planned, designed, constructed, operated and maintained by DMAT.

8.6.1 Objectives

The purpose road lighting infrastructure is to provide visibility on the road network at night. Road lighting's key focus is the safety and comfort of the road users, pedestrians and residents. In order to achieve this goal, road lighting must be provided in a high quality, consistent manner.

8.6.2 Key Design Considerations

The safety of all road users is the most important consideration in road design, and appropriate lighting is one of the key aspects ensuring a safe roadway. As such the design of a lighting system must consider a wide range of aspects as listed below:

- Driver Visibility requirements based on road category, travel speed, brightness hierarchy and balance with surrounding conditions;
- Appropriate road surface brightness uniformities, surround ratios and fixture glare limits;
- Conflict zones and High incident areas;
- Pedestrian and cyclist visibility;
- Coordination with the public realm and public realm lighting;
- Sustainability (e.g., use of LED's, reduced maintenance requirements and reduction of excessive lighting levels);
- Weather conditions (e.g., lighting coping with fog conditions);
- Driver comfort and fatigue avoidance;
- CPTED; (Crime Prevention through Environmental Design)
- Available technology;
- Emerging technology;
- Minimising lighting infrastructure;and
- Energy saving lighting management systems including off peak dimming profiles.

8.6.3 Applicable Standards and Guidelines

Road lighting infrastructure shall be planned, designed, constructed, operated and maintained in accordance with the following requirements, standards and guidelines:

Sector-Specific Documents

Lighting Manual	(PR-402)
Road Structures Design Manual	(TR-516)

Other References Documents

Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)
Public Realm Design Manual (PRDM)	(PR-401)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Urban Street Design Manual (USDm)	(ROW-603)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)

8.7 Public Lighting Infrastructure Standards

The objective of public lighting is to provide a safe and inviting public environment for the community within the hours of darkness. Public lighting can be used to practically light public areas as well as provide additional social, cultural, wellbeing, commercial and aesthetic value to the Public Realm.

8.7.1 Key Design Considerations

Public lighting needs to be designed to coordinate with the sector landscaping designs, as well as various areas and activities in the Public Realm. It is important that lighting is designed by someone with both theoretical and practical knowledge of lighting design and lighting technology. The following key design consideration should be taken into account when designing public lighting:

- Visual comfort, attractiveness, safety and light quality;
- CPTED; (Crime Prevention through Environmental Design);
- Purpose of Space and/or task;
- Activity and sports areas area to be illuminated ;
- Areas outside the activity space requiring illumination;



- Coordination with adjacent road lighting;
- Landscaping design (e.g., Coordination of lighting, planting and landscape design);
- Aesthetics of lighting as part of the area landscape furniture and design;
- Available technologies;
- Emerging technologies;
- Energy saving lighting management including off peak switching or dimming; and
- Minimizing of lighting infrastructure.

8.7.2 *Applicable Standards and Guidelines*

Public lighting infrastructure shall be planned, designed, constructed, operated and maintained and in accordance with the following requirements, standards and guidelines:

Sector-Specific Documents

Lighting Manual	(PR-402)
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Other References Documents

Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)
Public Realm Design Manual (PRDM)	(PR-401)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Urban Street Design Manual (USDm)	(ROW-603)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)

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9

GAS INFRASTRUCTURE



9.1 Overview

This section will provide a brief overview of the gas distribution infrastructure sector in the Emirate of Abu Dhabi. This will be done in accordance with the various sub-sectors that were defined in the Infrastructure Standards Evaluation Project; namely, Gas Distribution Infrastructure from high pressure off-take, up to and including end users appliances connections.



Figure 8: Natural Gas distribution PRS Facility

9.2 Context in Abu Dhabi

ADNOC Distribution, following Supreme Petroleum Council's mandate, is progressing the development of the Natural Gas (NG) downstream business in the Emirate of Abu Dhabi. This endeavor involves the development of the infrastructure as well as of the market, targeting to cater the residential, commercial, transportation, small and medium industrial and district cooling segments' demand.

The Project covers the design, construction, operation, maintenance, emergency activities and communication networks (SCADA, Fiber Optics) from / including the high pressure gas pipelines to the natural Gas Distribution Network terminating at Residential, Commercial and Industrial customers' appliances. The PROJECT will also serve the transportation sector by building and operating state of the art natural gas filling stations, vehicle conversion workshops and vehicle conversions.

The PROJECT Scope also includes customer acquisition, customer service, type approvals of gas meters/appliances and the design, construction and operation of a Gas Operations and Service Centre.

ADNOC Distribution currently operates networks at Yas Island, Al Raha Beach and Al Maryah Island and 31 Compressed Natural Gas (CNG) refilling stations, serving approximately 2,300 residential customers, 94 commercial customers , more than 5,500 vehicles and 4 conversion centers”

ADNOC Distribution will be responsible for the safety operation of the Gas Distribution Network and related infrastructure and services in the city and urban areas. The pipework and appliances inside the customers' premises is the responsibility of the customer and will be supplied, constructed and maintained by other private sector participants in the gas industry. However, ADNOC Distribution will determine and approve standards for equipment and installation within the customers' premises and the accreditation of private sector participants.*

* ADNOC DISTRIBUTION.

9.3 Gas Distribution Infrastructure Standards

9.3.1 Objectives

ADNOC Distribution has a clear mandate to develop Gas Distribution Network in Abu Dhabi Emirate, with holistic scope covering Natural Gas (NG), Compressed Natural Gas (CNG) for vehicles (NGV), and Synthetic Natural Gas (SNG) through underground distribution infrastructure networks. There is a clear focus on residential, commercial and customers, in addition to NGV, to provide the cleanest, safest, cost effective and reliable energy solutions.

Gas Distribution network infrastructure are included within the road and utility corridors right of ways, and relate to infrastructure services in this respect.



9.3.2 *Applicable Standards and Guidelines*

As indicated in the introduction, detailed standards and specification for gas infrastructure is regarded as a specialized industry. However, when such infrastructure is located within a road or utility corridor right of way, then the following standards shall apply:

Sector-Specific Documents

Material and Equipment Standards and Specifications	(GA-901)
Gas Distribution System - Construction Standards and Specifications - Polyethylene Distribution Mains and Gas Services	(GA-902)
Gas Distribution System - Construction Standards and Specifications - Steel Distribution Pipelines and Gas Services	(GA-903)
Design Basis Memorandum	(GA-904)

Other References Documents

Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)

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10

DISTRICT COOLING **INFRASTRUCTURE**



10.1 Overview

District cooling refers to the means of producing distributed and centralized cooling energy. The underlying process of district cooling is based delivering chilled water, created in a centralized and efficient cooling plant, through underground pipelines to many buildings that are located nearby.

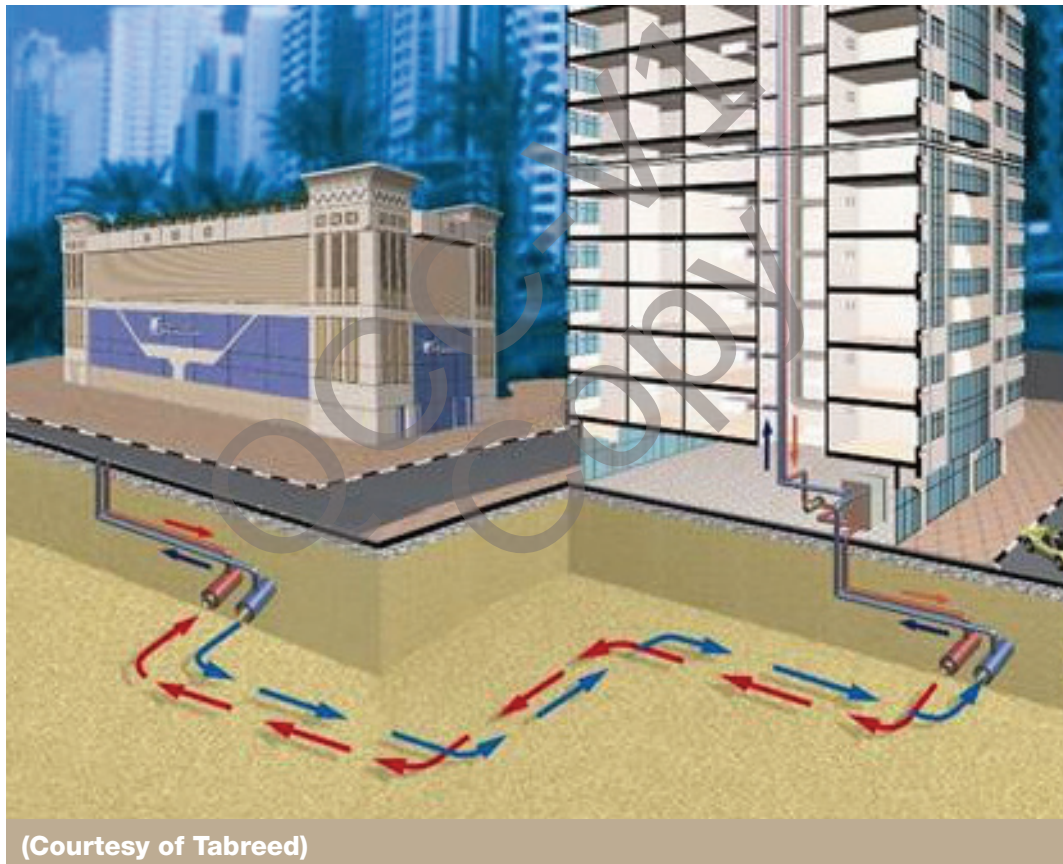


Figure 9: District Cooling Graphic

District cooling is a suitable alternative to traditional electric air conditioning systems because it typically consumes 40% to 50% less energy. These savings provide the opportunity for significant financial and environmental benefits.

10.2 Context in Abu Dhabi

Due to the desert climate of Abu Dhabi and to the high level of temperature and humidity in summer, electric-based air conditioning contributes to almost 70% of the Emirate's electricity consumption. By 2030, cooling demand in Abu Dhabi is expected to increase by nearly 300%, which creates a sense of urgency to move towards the widespread adoption of more energy efficient air conditioning systems like district cooling.

The characteristics of district cooling systems are complimentary to the urban development planned for Abu Dhabi. Specifically, it is suitable for high-density developments such as urban business districts, campuses of schools and universities, as well as residential complexes and towers. In Abu Dhabi, the National Central Cooling Company PJSC (Tabreed) is operating 39 district cooling plants, and has 69 district cooling plants in total across GCC countries.*

* TABREED COMPANY.

10.3 Objectives

The planning, design, construction, operation and maintenance of district cooling infrastructure through the Emirate of Abu Dhabi are carried out by companies, including National Central Cooling Company PJSC (Tabreed). Overall, the primary aim of Tabreed is to promote the production and distribution of chilled water from a central source for the purposes of air-conditioning.

This service plays a vital role in development of the Emirate of Abu Dhabi, considering the hot desert climate consistent with the region. District cooling serves as a sustainable alternative to traditional air conditioning systems, with a number of environmental and economic benefits.

District cooling infrastructure is included within road and utility corridor right of ways, and as such relate to infrastructure services in this respect.

10.4 Key Design Considerations

District cooling needs to be designed in coordination with the designs of complimentary utilities like water and electricity. It is important that district cooling is designed by someone with both theoretical and practical knowledge of the functions and technology of centralized cooling systems. The following key design consideration should be taken into account when designing district cooling systems:

- Efficiency;
- Sustainability;
- Reliability;
- Safety;
- Complimentary utilities;
- Proximity to end use; and
- Thermal comfort.



10.5 Applicable Standards and Guidelines

As indicated in the introduction, detailed standards and specification for district cooling is regarded as a specialized industry and is therefore not referenced. However, when such infrastructure is located within a road or utility corridor right of way, then the following standards shall apply:

Sector-Specific Documents

A Comprehensive regulatory framework is being developed and a reference will be provided once approved.

Other References Documents

Roadside Design Guide	(TR-518)
Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)

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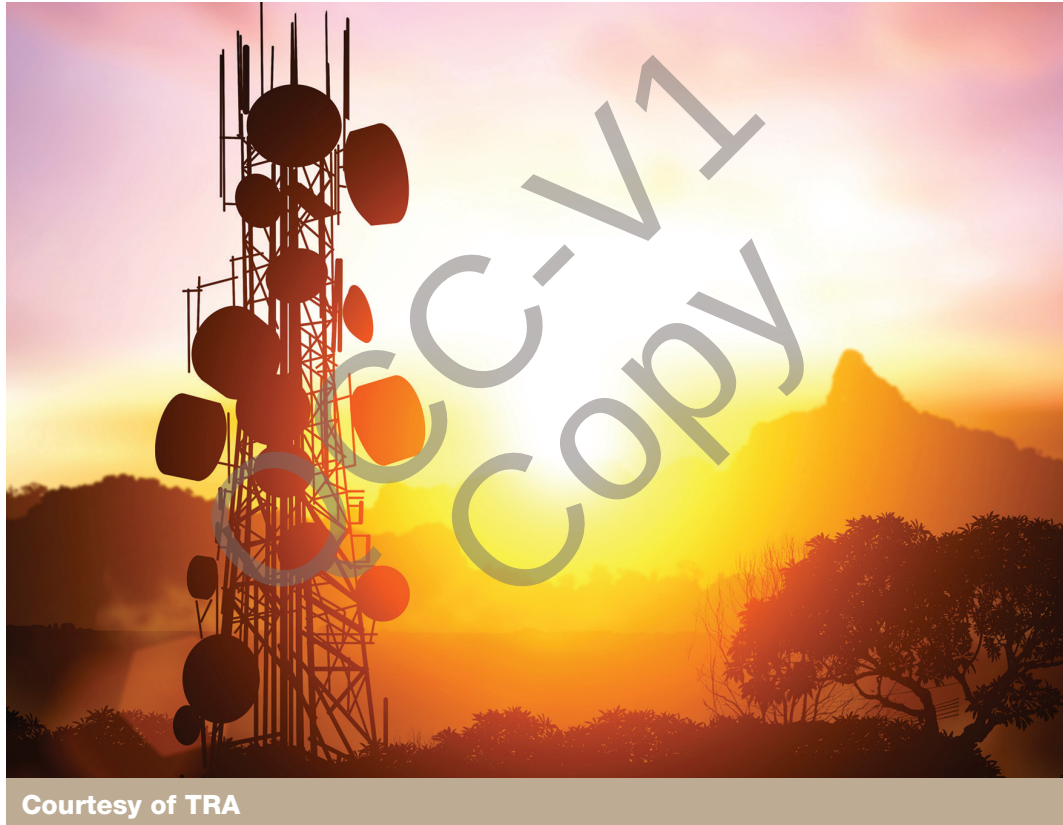
11

Telecommunications **Infrastructure**



11.1 Overview

Modern telecommunications has become an essential aspect of life for many communities and has been continuously growing to contribute to the growth of the national economy.



Courtesy of TRA

Figure 10: Telecom Tower

The rapid pace of development in telecoms has led to new communications technologies constantly being upgraded to improve services – for homes, businesses, public utilities and electronic commerce. This document provides the guideline for the design requirements, proper ways of implementation and technology specifications of public infrastructures being installed in fixed networks, as well as the common standards and specifications for mobile site construction including in-building environments. This document is applicable for telecommunications services being carried out by national telecoms operators in the emirate of Abu Dhabi.*

* TRA: Telecommunications Regulatory Authority.

11.2 Applicable Standards and Guidelines

Sector-Specific Documents

Telecom Civil Infrastructure Guidelines for Fixed Networks - Design Requirements	(TE-1101)
Telecom Civil Infrastructure Guidelines for Fixed Networks - Implementation and Specifications	(TE-1102)
Common Standards and Specifications on Civil Works for Mobile Network Site Construction	(TE-1103)
Common In-Building Specification Guideline	(TE-1104)

Other References Documents

Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)





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12

WASTE MANAGEMENT **INFRASTRUCTURE**



12.1 Overview

The planning, design, construction, operation and maintenance of waste management infrastructure through the Emirate of Abu Dhabi are carried out by the Center of Waste Management, Abu Dhabi (Tadweer). Waste management infrastructure is included within the roadside environment, and therefore relates to public infrastructure in this respect. This section discusses infrastructure relating to the permitting, collection and treatment of solid and hazardous waste. Additionally, it includes lists of guidelines and regulatory requirements related to management of waste including, non-hazardous, hazardous waste as well as medical waste.



Courtesy of CWM (Tadweer)

Figure 11: C + D Waste Recycling.

However, it does not include standards guidance for the construction of landfills or other waste collection or disposal facilities.

12.2 Context in Abu Dhabi

As part of the Abu Dhabi Vision 2030, the Emirate aims to be amongst the leading countries in waste management. To achieve this goal, there is a pressing need to invest in waste management infrastructure ranging from collection, treatment, recycling and disposal.

In 2015, the total amount of waste generated in the Emirate of Abu Dhabi reached approximately 8.39 million tons per annum, with a rate of 22,993 tons per day. From the total amount of waste generated, municipal solid waste accounted for

approximately 20%, and reached about 1.67 million tons per annum in 2015, as well as the construction and demolition waste accounted for about 34% of the total waste in 2015.

Segregation of recyclables and waste as per the waste classification, as well as segregated waste collection is key to the sustainable waste management process, and is one of the challenges facing the Abu Dhabi Emirate. Abu Dhabi has distributed more than 140,000 green and black bins for segregation of domestic waste into recyclables and putrescible. By early 2017 the residential collection system will be supported by collection points within the residential areas in Abu Dhabi and Al Ain regions to facilitate residents to deposit recyclables like paper & cardboard, plastic, metals, glass, used cooking oil, domestic hazardous waste, etc. As Abu Dhabi continues to develop this, and other waste management systems, it will provide a series of centralized and regulated collection facilities with the aim of preventing unauthorized dumping. The Emirate of Abu Dhabi has achieved a major milestone by using recycled concrete aggregates as road base material and in other infrastructure projects like Etihad Rail. Figure1 shows the construction and demolition waste recycling facility in Abu Dhabi.*

* CWM: Center of Waste Management (Tadweer) .

12.3 Applicable Policies, Standards and Guidelines

Applicable Policies, standards and guidelines relating to waste management infrastructure with respect to public infrastructure are as follows:

Sector-Specific Documents

Waste Classification Policy	(WM-1201)
Waste Planning Policy	(WM-1202)
Licensing and Enforcement Policy for Waste Sector	(WM-1203)
Waste Collection, Segregation, Transfer and Tracking Policy	(WM-1204)
Waste Classification Technical Guideline	(WM-1205)
Waste Reuse, Recycling, Resource Recovery, Waste to Energy, Treatment and Disposal Policy	(WM-1206)
Certification Scheme for Recycled Crushed Aggregate	(WM-1207)
Certification Scheme for Re-refined Engine Lubricant Base Oil	(WM-1208)
Certification Scheme for Biodiesel	(WM-1209)
Requirements and Procedures for the Disposal of Hazardous Wastes (CWMTG 2# merged)	(WM-1210)
Requirements and Procedures for Registration of Skips and Container	(WM-1211)
Registration of Waste Management Consultancy Offices	(WM-1212)



Inspection of Waste Treatment Facilities, Waste Transportation and Pest Control Facilities	(WM-1213)
Permits and Licensing Requirements for Transportation, Treatment and Recycling Facility	(WM-1214)
Management of Asbestos / Asbestos Containing Material (ACM)	(WM-1215)
Estidama Pearl Building Rating System: Design and Construction	(DP-306)
Estidama Pearl Community Rating System: Design and Construction	(DP -307)
Estidama Pearl Villa Rating System: Design and Construction	(DP -308)

Other References Documents

Standard Operating Procedure for Permitting of Development and Infrastructure Projects	(TR-519)
Work Zone Traffic Management Manual	(TR-531)
Right of Way Utility Distribution Manual (for utilities in non-urban areas)	(ROW-601)
Utility Corridors Design Manual (UCDM, for utilities in urban areas)	(ROW-602)
Utility Plot Design Manual (UPDM)	(ROW-604)
Standard Conditions for Working in Proximity to Water Transmission Assets	(WA-707)
Standard Conditions for Working in Proximity to Power Transmission Assets	(EN-805)
Executive Council, Decree No. BT 9G 25/2010 dated 26th July 2010, requiring the mandatory use (subject to availability) of a minimum of 40% aggregate (by volume) of recycled construction and demolition waste in roads projects, as well as any other infrastructure projects throughout the Emirate of Abu Dhabi	Decree No. BT 9G 25/2010

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13

NOC PROGRAM FOR THE UTILITIES AND INFRASTRUCTURES IN THE **EMIRATE OF ABU DHABI**



The following section aims to guide consultants and contractors in utilizing the Municipal Approvals Services Platform “One-Stop Shop” / Common NOC Solution for obtaining the required permits, approvals, and No Objection Certificates (NOCs) pertaining to the various building, constructions, and utility and infrastructure projects and activates.

This section, briefly yet sufficiently, provides information on the various requirements in terms of standardized inputs and outputs towards successful processing and completion of transactions, and points readers to published resources for further information.

13.1 Introduction to No Objection Certificates (NOCs)

A No-Objection-Certificate (NOC) is an official document issued by a utilities and infrastructure asset custodian, in the process of responding to an application from a requester (government and semi-government entities, developers, contractors, consultants, or individuals) that intends designing or executing a project or performing fieldwork in a specific location. Through granting a NOC, the asset custodian confirms the lack of impact (or possibility of mitigation of any probable impact) by that project throughout its various stages of development, on the integrity, utilization, and performance of the assets. The owner of the specific development project (or their representative) is responsible for acquiring and renewing NOCs as necessary.

No Objection Certificates are compulsory and are pre-requisite to satisfy the requirements of approvals and permits and their issuance by the municipal sector of Abu Dhabi Emirate. The approvals and permits transactions that necessitate NOCs are categorized into the following major groups within the overall long-term urban planning and development:

- Plot Services: Transactions that relate to plots but do not involve construction works, such as land subdivision, merging, and extension.
- Buildings and Constructions: Transactions that relate to permits for building and other construction activities that take place within a plot.
- Utilities and Infrastructure: Transactions that relate to activities that take place within the public domain, which relate mainly to the planning.

Approvals and permits (and accompanying NOCs) are required throughout the “asset” management lifecycle of utilities and infrastructure, buildings and constructions, and land properties, comprising the four stages of planning and design, implementation, operations, and decommissioning.

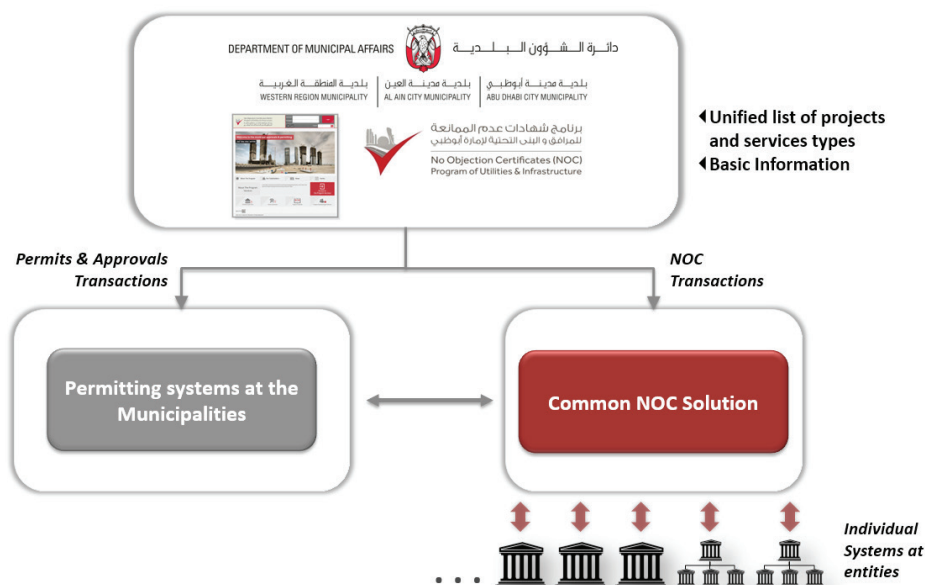
13.2 The Municipal Approvals Services Platform “One-Stop Shop”

The Municipal Approvals Services Platform “One-Stop Shop” (<https://noc.abudhabi.ae>) provides common platform for submitting applications for approvals, permits, and NOCs pertaining to building, constructions, and utilities and infrastructure development projects.

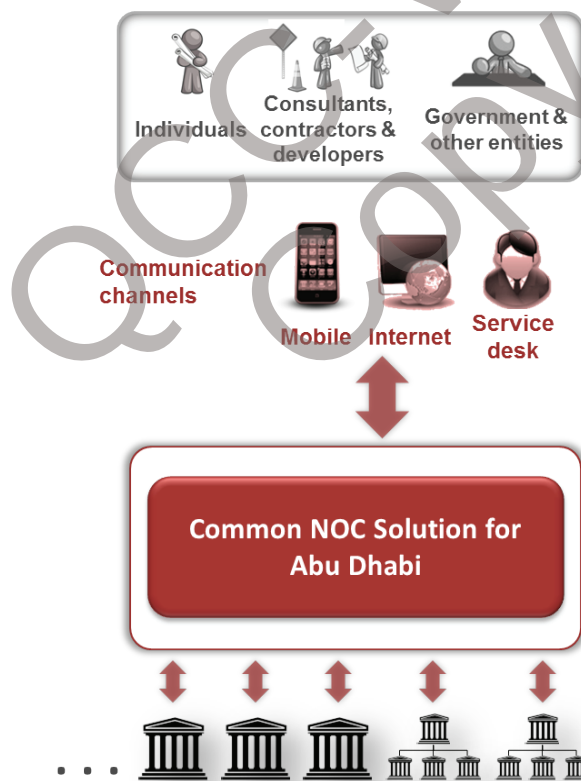
With its common interface across the three municipalities and between the municipalities and the NOC stakeholder entities, the solution provides the following features and merits:

- A unified window that serves all types of applicants and requesting entities (citizens, consultants, contractors, developers, and government & semi-government entities).
- A unified and comprehensive list of projects/activities that require permits, approvals, and NOCs that guides the applicant in the submission process.
- A comprehensive set of main and support services
- Common and specific/tailored workflows.
- Detailed and entity-specific configurable business rules that govern the behavior of the system at every step.

With submitting a common application (with all supporting information and attachments), an applicant will be able on the one hand to satisfy all NOC requirements of the various NOC issuing entities (asset custodians), and on the other hand to carry on the process of acquiring the necessary permit or approval from the relevant permitting system in the concerned municipality.



With regard to NOCs, the Municipal Approvals Services Platform “One-Stop Shop” aims to streamline the processing of NOC transactions that are solicited by individuals or entities as a requirement for obtaining necessary approvals and permits. This is achieved via the solution through determining all the NOC input requirements as per the sought permit or approval, and then allowing the applicant to obtain, in a simplified manner, all required NOCs from all concerned stakeholders in one single transaction through various communication channels that are subject to continuous development and improvement. The following diagram illustrates the “One-Stop Shop” for NOC Services through the Common NOC Solution:



The Common NOC Solution facilitates completing the submittal process online (supported in specific cases by direct interaction as may be requested by the municipality or the NOC issuing entities). The Common NOC Solution automates the administrative process, optimizes the points of contact, and streamlines data submittals for consultants and contractors (and other applicants).

The Common NOC Solution provides, using a one-stop-shop portal, information for all types of requests providing a complete package on submittal requirements, the process, fees, outputs, durations, and contacts for the various NOC transactions.

The NOC one-stop-shop aims to improve interagency collaboration, establish minimal points of customer contact, and make collective improvements to the overall customer process (through streamlined/standardized requirements, processes, and outputs).

13.3 NOC Requirements in the Context of Approvals and Permits

This section aims at describing the NOC requirements (in the context of approvals and permits) as per the major stages of the asset management lifecycle for land properties, buildings and constructions, and utilities and infrastructure.

13.4 NOCs as Mandatory Requirements for Approvals and Permits

NOCs are mandatory for certain land administration transactions in two key situations:

- The delineation of land plots pertaining to the allocation or re-allocation (in some cases in the transfer of ownership) with the objective of issuing the final approval of land plots drawings, noting that the initial approval may require in certain cases a “Request for Information” from the concerned entities.
- Transactions that incur changes on the dimensions of land plots including plot extension, division, merging (and corridor merging), and adjustments on plot boundary.

The lifecycle of buildings and other constructions necessitates NOCs at the following stages:

- The conceptual approvals for construction projects, and then the construction stage (including temporary fencing, when applicable).
- Transactions that result in changes on the building surface or sub-surface footprint (e.g., building or basement extension).
- Demolition.

For utilities and infrastructure, NOCs are necessary at the following cases:

- The planning and design stage.
- Field works associated with the construction/installation, and as necessary during the maintenance and decommissioning of assets.



The following table summarizes the NOC requirements as per the various stages of the asset management lifecycle:

V

Permit/Approval Type → Stage ↓	Land Administration	Buildings & Constructions	Utilities & Infrastructure
Planning / Design	✓		✓
Implementation (Construction / Implementation/ Completion / Commissioning / Allocation)	✓ Allocation	✓ Design & Construction	✓ including the construction/installation of utilities as well as all maintenance operations that involve field work (whether or not they necessitate replacement or rerouting of utility lines), and the decommissioning of utilities that require field works
Operation and Maintenance / Modification	✓ Plot extension, division, merging, correction/shifting, etc.	✓ Building extension / modifications	
Retirement (Decommissioning / Demolition / Uninstallation)		✓ Demolition	

✓ NOCs Requirements

13.5 NOC Inputs

Inputs necessary for NOC processing and operations are usually entered directly through the Common NOC Solution. The Common NOC Solution can also receive the NOC inputs through the electronic forms of the permitting and approvals systems upon the readiness of such solutions, and according to the approvals and permits framework in municipalities that identifies a wide range of inputs as requirements for approvals and permits applications. The Common NOC Solution can also acquire the necessary inputs for NOC processes through integrating with other shared government services and databases.

The Common NOC Solution is designed to accommodate all the inputs that are required for processing NOC requests. Each type of NOC transaction has its own specific requirements in terms of inputs, and this is inherently linked to the original permit or approval transaction that triggers the NOC request.

The NOC inputs are broadly streamlined and optimized into the following groups as per the requirements of the Common NOC Solution:

- Attributes or submittals concerning the identity of the requester, their credentials, and other administrative aspects;
- Attributes or submittals that cover the technical aspects of the proposed works;
- Attributes or submittals that describe the location and spatial characteristics of the proposed works.

The following table provides a general outline of the identified NOC input requirements for each main group, and provides a brief description of those requirements:

ADMINISTRATIVE INPUTS	
Applicant Identification	Information of the Applicant: entity name (consultant/ contractor), actual applicant, contact information
Letter of Appointment / Assignment / Authorization	Letter from entity of which the Applicant is a representative
Commercial Registration	Authentication of the commercial registration (which could be consumed as a shared government service from the Department of Economic Development) in case of commercial entity applicants
Administrative Attributes	Attribute information that hold administrative characteristics or specifications of the Applicant
License	Any license other than the general commercial registration
Power of Attorney	A written authorization to represent or act on another's behalf in matters relating to the transactions
Certificate	Any certificate such as Health and Safety Certificates, and consultants and contractors classification certificates
Letter of Undertaking	Letter from the Applicant committing to comply with required conditions



Permit / Approval (as per stages)	A prerequisite approval from a government entity or asset owner
Site Handover (after demarcation)	Letter from municipality confirming handover of a site or parcel
Fees	Applicable fees against the service of receiving and processing of NOC application and the issuance of NOCs (Refer to section 3.2.4 in this document)
Bank Guarantee	Issued by a bank to ensure that a party (contractor/consultant) fulfills its contractual obligations
Correspondence	Previous correspondence
Other support documents	Any other type of administrative document
TECHNICAL INPUTS	
Project Characteristics	Attribute information that hold technical characteristics or specifications of the project
Project Schedule	Schedule of works, including the reinstatement schedule
Method Statement	How the work will be conducted, especially relevant when street excavations or tunneling is to be performed, including as applicable a relocation/ protection plan with regard to existing utilities
Design Criteria	Design information about the project to be executed, which could include materials to be used, reinforcement, protection measures, etc.
Risk Assessment	Assessment of risk that may be required by certain NOC Issuing Entities
HSE Plan	Health, Safety, and Environment Plans, as required
Traffic Impact Study	As required by DMAT, or Abu Dhabi Police
Test Results	Testing of ground surface and subsurface conditions, such as soil analysis / compaction tests, geotechnical, geophysical, etc.
Photographs	Any photographs of the site (Pre-construction, as-found, or completed works)
Other Technical	e.g., Technical Reports, Excavation / Field Survey reports, road furniture inventory lists, etc.

SPATIAL INPUTS (Comprises format, type, map scale, unified spatial reference, basemap, cadastral areas, and other parameters as applicable)	
Site Plan	Plan of the specific parcel issued by the municipality
Location Map	Plan of the site prepared by the applicant based on common spatial data, and approved by the municipality
Affection Plan	Official site plot plan issued by the government with survey coordinates delineating the boundary. An Affection plan will provide details on the total plot size, the allowed built-up area, and so forth
Route Approval	Approved route issued by the municipality for utility and infrastructure service lines projects
Location Attributes	Attribute information that hold location / spatial characteristics or specifications of the project
Project Location	Drawing of the proposed works (project boundary + intended works) indicating excavations and other activities that may impact assets
Cross Section	Drawing showing the cross section to indicate the location and depth of proposed works
Field survey	Field survey results, including as-built drawings, for executed works or existing site conditions for site investigation
Other Drawings and Spatial Information	e.g., MEP Plans, Structural Drawings / Model / Calculations, Architectural Drawings, spot height and leveling sheets, etc.

13.5.1 Spatial Data

NOCs (and related processes) are characteristically a “spatial” subject as they aim at safeguarding the integrity, performance, and efficiency of existing and planned utilities and infrastructure at the geographical vicinity of intended works. Accordingly, the spatial characteristics and dimensions of such works as well as of neighboring utilities and infrastructure assets are prime determining factors of NOC requirements and resultant NOC outputs. It is hence that spatial data play a significant role in the processing of NOCs for all types of permitting and approval transactions and throughout the various stages of asset management lifecycle.



It is important to mention that the Common NOC solution provides different access channels to enable authorized users to view, query, utilize, physically download (in specific cases and controlled manner), and upload basic spatial data layers during the submitting and processing NOC of transactions in the Common NOC Solution at the various different stages of development projects lifecycle.

Request for Information (RFI)

The Common NOC Solution facilitates and simplifies the process of consultants and contractors acquiring the necessary information for their projects, and reduces the number of Requests for Information (RFIs) by giving the consultants/contractors access to specific information within their projects boundaries.

The Request for Information service (for planning and main coordination purposes) aims to provide the concerned stakeholders through the Common NOC Solution with the spatial information necessary for the planning and design of major and detailed development projects. The access to common spatial information by consultants and contractors for internal use of projects' owners will help to optimize projects design and reduce the possibility of conflicting with other assets in later stages of project development lifecycle.

On the transactional level, the Common NOC Solution enables NOC applicants to view (limited within a geographical and temporal window) the spatial information pertaining to the concerned project area to support the preparation of the NOC application. This satisfies in an automated and transactional fashion what was known as "Request for Information" (RFI) or "Notice of Intent" (NoI) in the procedures that were adopted before the operation of the Common NOC Solution).

In general, access mechanisms to information pertaining to stakeholders are governed by the data sharing rules of those entities and according to the geographical areas and types of assets.

Spatial Data Submission Framework

Data pertaining to building, constructions, and utility and infrastructure development projects originates mainly from developers, contractors, and consultants. A key requirement, therefore, is to streamline the process of data submission by these bodies to the government entities by developing and implementing a spatial data submission framework. The NOC Program, in consultation with the relevant stakeholders, has developed and published NOC unified spatial data submission standards for utility and infrastructure projects (under the realm Permitting & Approvals).

Through these standards, which define Geospatial Data Submission (GDS) specifications for spatial features and attributes in GIS and CAD formats, consultants and contractors are provided with the basemap reference data for their planning work. Later, at the end of the project, consultants and contractors can provide the as-built data accurately referenced to the basemap reference data.

The standards provide a framework for the NOC stakeholder entities and data submitters to streamline data utilization, collection, and submission processes. These standards serve several purposes including enabling data custodian entities to receive surveying and planning data from consultants and contractors in a standardized format, in addition to improving data QA/QC and facilitating data exchange across the community of stakeholders.

Joint Field Survey Services

This service enables the requesting of field survey and excavation services on assets for particular area, which is executed through approved third-party service providers that possess sufficient experience are well equipped with advanced technological tools and therefore capable of providing standard services in this domain. Such service is provided under the supervision and guidance of the Department of Municipal Affairs and Transport as the governance body for these services, and within a framework that is in harmony with related policies and regulations.

In general, the need for such type of services arises in various situations within the different stages of implementing development projects. They could also be utilized for the final checking and approval of as-built drawings once projects are concluded and the concerned consultant reaches the “Project Completion Certificate” stage. The applicant (i.e., the consultant) submits through this service their request coupled with all necessary information and as-built drawings in digital format and in accordance with required standards and criteria. The request is then reviewed and approved by the requesting entity of the service (i.e., the project owner) as well as by the projects representatives of the concerned municipality. After that, the field survey and excavation officer of that municipality will issue a “task order” with all specifics and requirements to the third-party service provider while notifying all entities that have issued NOCs for that particular project with specific conditions and requirements. The service provider will then conduct the requested field survey and excavation activities and deliver the as-built data as verified on the ground. Such data needs to obtain the final approval of the requesting consultant followed by the concerned municipality, and the requesting entity (the project owner) will then be notified of the final “as-built” approval.



13.5.2 NOC Fees

With regard to service fees related to NOCs operations, the NOC Program periodically examines as necessary the existing situation pertaining to fees on the basis of analytical study of the financial and other aspects, in the light of related laws and regulations, and with taking into consideration the local and international best practices in this regard. An e-payment online facility (via the Common NOC solution) is currently under development to enable the electronic payment of all NOC service fees as part of completing the NOC application process. The envisaged solution will be availed progressively during the coming stages of operations upon administrative and technical readiness by the various concerned stakeholder entities.

Regarding the currently applicable practices for payment of NOC service fees, the Common NOC Solution identifies the due fees to all concerned stakeholder entities based on the business rules agreed upon with each entity and configured accordingly in the system. Applicants, who are capable of viewing the due fees and print a hardcopy list for their follow-up and pursuance, shall pay the required fees at the earliest through the channels, and in accordance with the procedures and requirements, stipulated by these entities. Applicants are also provided with the necessary tool and guidance in the system to upload payment receipts, which are packaged with NOC transactions received electronically by entities through the system. Subsequent communication and follow-up on fees payment between the entity and the applicant is to take place in accordance with the entities' practices and procedures.

13.6 NOC Outputs

NOC outputs (which are key inputs towards issuing the final municipal approvals) are the “responses” of NOC issuing entities (i.e., utilities and infrastructure assets custodians) to NOC applications as outlined below, which are supplemented with a set of documents and information/data as applicable:

- **“NOC with No Conditions”**: Approval (No objection) issued by the asset custodian entity after reviewing the specific transaction and identifying no substantial impact from the intended works on their assets. Such approval could also be automatically issued as a “default” NOC based on specific agreed-upon business rules between the permitting body and the NOC issuing entities.
- **“NOC with Conditions”**: An approval associated with instructions and requirements specific to the issuing entity such as HSE measures and requirements, work permits, or additional attachments to be provided such as method statement report or any other necessary studies.
- **“Conditional NOC”**: A pending approval tied to conditions or specific requirements that could necessitate additional works or efforts on the implementing party due to potential impact on assets. The conditional NOC leads the requesting entity of the NOC to the following scenarios

according to their agreement with the custodian entity of the assets subject to potential impacts:

- Modifying or redesigning the intended project.
 - Implementing additional safeguarding measures (that could involve additional costs on the contractor or the project owner) according to an agreed-upon methodology statement.
 - Relocating / re-routing of the assets of potential conflict with the intended project
- **“Objection”**: An objection typically results from identified substantial impact of the intended works on the assets of the custodian entity (or due to the lack of sufficient information and clarifications from the requesting entity).
 - It is important to emphasize that the intent of government entities is to avoid “objection” to NOC requests to the extent practical. To achieve this target, emphasis needs to be given on the planning stage that is developed using best available information.

In the Common NOC Solution, the permitting body (represented typically the town planning sector in the municipality) issues a single common and consolidated NOC output, which comprises the individual NOC outputs of the various asset custodians, and constitutes a significant element in processing and closing approvals and permits transactions.

13.7 The NOC Emergency Management Framework

The NOC Program, as emphasized in the NOC Program Policy, has recommended the development of a joint governance framework among the stakeholder entities for emergency management and reporting that would allow them to leverage the Common NOC Solution for promptly sharing information about confirmed incidents on their respective assets. This intends to provide the means for collaboration among the impacted parties while the emergency situation is being contained during the transition period until an emergency response plan is developed and disseminated to all stakeholders.

The Common NOC Solution demands all NOC requesters to follow the standard workflows and procedures as established by the NOC Program. The Common NOC Solution does however accommodate valid exceptions, which shall specifically relate to emergency situations that involve damages to assets in the public realm, injuries to individuals, or disruption of services. Those emergency situations, with their imminent threats or dangers to life, health, property, utilities and infrastructure, or the environment, require urgent intervention and exceptional actions in order to avoid further exacerbation of the situation and assure continuity of public utilities and related services.

The emergency management framework of the NOC Program intends to satisfy the need for urgent works (and necessary work permissions) in the initial incident



containment stages and in some cases for further works towards full recovery. This is achieved through forming and enhancing a common practice between the concerned stakeholders for responding to and managing such emergency incidents through coordination between the impacted entity and other stakeholders who may have exiting assets in the incident location.

Further Information:

For assistance or further information pertaining to utilizing the Municipal Approvals Services Platform (One-Stop Shop) / Common NOC Solution, please communicate with the NOC Program support team via the following contact information:

- Phone: +971 2 403 0378 (Sunday to Thursday - 8:00 am to 4:00 pm)
- Email: noc.helpdesk@adsic.abudhabi.ae

In addition, the following link provides online access to the Municipal Approvals Services Platform (One-Stop Shop) and the Common NOC Solution <https://noc.abudhabi.ae>. The links titled “Program Guidelines” and “Corporate Support Services” on the main page avail the latest published material necessary for applicants to submit their NOC applications successfully, including:

- NOC Guidelines
- Instructions and awareness material
- Training material
- The unified list of projects/activities that require permits, approvals, and NOCs
- CAD/GIS templates
- NOC unified spatial data submission standards
- Specific additional requirements by NOC issuing entities

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Appendix 1:

ACRONYMS AND **GLOSSARY OF TERMS**





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1 List of Acronyms

AADC	Al Ain Distribution Company
ADDC	Abu Dhabi Distribution Company
ADM	Abu Dhabi City Municipality
ADNOC	Abu Dhabi National Oil Company
ADSSC	Abu Dhabi Sewerage Services Company
ADWEA	Abu Dhabi Water and Electricity Authority
ADWEC	Abu Dhabi Water and Electricity Company
AAM	Al Ain City Municipality
ACM	Asbestos Containing Material
CWM (Tadweer)	The Center of Waste Management Abu Dhabi - Tadweer
DMAT	Department of Municipal Affairs and Transport
EAD	Environment Agency Abu Dhabi
GCC	Gulf Cooperation Council
GDCCD	General Directorate of Abu Dhabi Civil Defence
GWh	Gigawatt hours
km	Kilometre
kV	Kilovolt
LDC	Load Dispatch Centre
m³	Cubic meters
MED	Multiple Effect Distillation
MIG	Million Imperial Gallon
MIGD	Million Imperial Gallon Per Day
MOI	Ministry of Interior
MWh	Megawatt hour
MSF	Multi-Stage Flash Desalination

NOC	No Objection Certificate
NOI	Notice of Intent
PRS	Pressure Regulating Station
QA/QC	Quality Assurance and Quality Control
RSB	Regulation & Supervision Bureau
RO	Reverse Osmosis
ROW	Right of Way
SCAD	Statistic Center Abu Dhabi
Tabreed	National Central Cooling Company
TRA	Telecommunications Regulatory Authority
TRANSCO	Abu Dhabi Transmission and Despatch Company
TSE	Treated Sewage Effluent
UAE	United Arab Emirates
UPC	Abu Dhabi Urban Planning Council
WRM	Western Region Municipality



2. Glossary of Terms

Community Facility: A facility that is used by and/or intended to support the residents of an area. The types include government and institutional facilities such as police, civil defence and postal services; religious facilities such as mosques; educational facilities such as schools, colleges and universities; social and recreational facilities such as wedding halls and cultural centres; and healthcare facilities such as clinics ⁽¹⁾

Competent Authority: means the entity that is required to approve the technical aspects of the projects and the design. ⁽²⁾

Developer: means an entity engaged in the preparation / development of a site for residential or commercial use. ⁽²⁾

Project Owner: means an entity that is or will be the owner of the infrastructure assets for which a permit is sought. ⁽²⁾

Right of Way (ROW): A corridor (located between plot boundaries) that is used by the public for all modes of transportation and utilities. ⁽¹⁾

Street: A public or private right-of-way, other than a major thoroughfare (e.g. highway) or a pedestrian path (e.g. sikka), that is used to carry vehicular traffic and pedestrians from one location to another, and/or provide vehicular and pedestrian access to an adjacent property. ⁽¹⁾

The Public Realm: Include all exterior places, linkages and built from element that are physically and/ or visually accessible regardless of ownership. These element can include, but are not limited to, streets, pedestrian ways, bikeways, bridges, plazas, nodes, squares, transportation hubs, gateways, parks, waterfronts, natural features, view corridors, landmarks and building interfaces. ⁽¹⁾

“Urban Street”: means a road within an urban area serving both vehicles and the vulnerable users (pedestrians and cyclists). ⁽¹⁾

Utility Corridor: A swathe of land or right-of-way that is intended for the infrastructure and facilities that are related to the supply and conveyance of utilities. ⁽¹⁾

References

¹ Abu Dhabi Urban Planning Council

² Guide To Abu Dhabi Infrastructure Standards, Edition May 2012

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Appendix 2:

INFRASTRUCTURE PROVISION IN
THE EMIRATE OF ABU DHABI





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1 Department of Municipal Affairs and Transport

The Department of Municipal Affairs and Transport has been formed in November 2016 pursuant to a law issued by HH Sheikh Khalifa bin Zayed Al Nahyan, President of the UAE, thus replacing the ex “Department of Municipal Affairs” and “Department of Transport”.

Responsibilities of the Department of Municipal Affairs & Transport span a wide array of sectors including municipal affairs, civil aviation as well as land & marine transport sectors. The Department drafts legislations of relevance to these fields and submits them to the Executive Council. It also prepares relevant plans & strategies in accordance with the development policy of the government.

The Law states that the Abu Dhabi City Municipality, Al Ain City Municipality and the Western Region Municipality as well as any municipality that may be established in future, has to be affiliated to the Department in order to ensure their compliance with the applicable legislations, and ensure the sound implementation thereof. Such changes are part of the endeavours of the Department of Municipal Affairs & Transport to set a platform conducive to the improvement of living standards and capable of serving the entire needs of community members besides adding to their happiness by enhancing the services offered to them. At the same time, these responsibilities will support the government efforts towards achieving a balanced, comprehensive and sustainable development in the Emirate.

2 Abu Dhabi Water and Electricity Authority



2.1 Abu Dhabi Water and Electricity Authority (ADWEA)

Abu Dhabi Water & Electricity Authority is a national corporate wholly owned by the Government of Abu Dhabi maintaining legal & administrative independence. ADWEA is responsible for implementing government policy regarding water and electricity sector. ADWEA supplies Water & Electricity to a population more than two millions inhabitants through its subsidiaries companies.

In 1998, ADWEA nature has been changed by establishing special partnership with the most famous global consortiums, In order to implement the most updated types of technologies that used for production of power and desalination of water. Targeting to provide best services for consumers. ADWEA programs and initiatives



aim to achieve sustainability and planning for development. In observation to the environment and economic aspects. In order that to supply Water and Electricity to Abu Dhabi population through an effective and efficient means. To achieve the best results, optimizing responsibility and improving standard of services. ADWEA resorts to the privatization of the Water and Electricity Sectors.



شركة أبوظبي للتوزيع
Abu Dhabi Distribution Co.

2.2 ADDC

Abu Dhabi Distribution Company (ADDC), an entity of Abu Dhabi Water and Electricity Authority (ADWEA), incorporated in 1998, is responsible for distributing drinking water and electricity supplies to customers in Abu Dhabi city and its dependencies.

ADDC's customer base is in excess of 600,000 customers, for water supply and electricity distribution networks of 33Kv and 11Kv, residing in the company's three regions, which are Abu Dhabi Island, Eastern Region that includes Mussaffah, Bani Yas, Al Shahama and Al Khatem, the Western Region that includes Al Mirfa, Madinat Zayed, Gyathy, Liwa, Al Silaa and Delma Island.



شركة العين للتوزيع
Al Ain Distribution Company

2.3 AADC

Al Ain Distribution Company (AADC) is a public joint stock company registered and incorporated in the United Arab Emirates under Law No 2 of 1998 on 1 January 1999 with Head Office at Al Ain. AADC is a wholly owned subsidiary of Abu Dhabi Water and Electricity Authority with a fully paid capital of AED Billion 1,530,000,000.

AADC is the sole distributor and supplier of water and electricity in the Eastern Region of the Emirate of Abu Dhabi (Al Ain city and its surrounding rural areas). This includes the ownership, operation and maintenance of the water and electricity distribution network assets, meter reading, and services for the supply of water and electricity.

2.4 TRANSCO

Abu Dhabi Transmission and Despatch Company (TRANSCO) is a subsidiary of Abu Dhabi Water and Electricity Authority (ADWEA). TRANSCO's role consists of the planning, construction and operation of safe, reliable and efficient bulk water transmission and high voltage power transmission networks within Abu Dhabi and the Northern Emirates. TRANSCO is in charge of transmitting power and water from the Independent Water & Power Producers (IWPPs) to Abu Dhabi Distribution Company (ADDC) and Al Ain Distribution Company (AADC) as well as other distribution companies in northern emirates.

Abu Dhabi Transmission and Despatch Company also contributes to the operation of the Emirates National Grid. On the regional stage, TRANSCO contributes to the development of the wider GCC transmission grid, which will allow the trading and transmission of power across a number of countries.

2.5 ADWEC

The Abu Dhabi Water and Electricity Company is a wholly owned subsidiary of the Abu Dhabi Water and Electricity Authority (ADWEA) and is the Single Buyer and Seller of Water & Electricity” in the Emirate of Abu Dhabi. ADWEC supplies water and Electricity to consumers through long term analysis of consumption patterns, while planning to increase production throw power and water purchase agreements with the generation companies, and through sales contracts on the basis of a Bulk Supply tariff (BST) with the distribution companies. ADWEC also purchases fuels(natural gas) from fuel suppliers for the producers.

2.6 Al Mirfa Power Company (AMPC)

On January1st 1999, Al Mirfa Energy Company was established as a joint venture corporate comprised of Al Mirfa Central Plant and Madinat Zayed Power Plant with a maximum capacity of 11 megawatts per hour. In 1993, work started on Al Mirfa Central Plant to improve services to customers in the Western Sector. The actual production of the Plant began in 1995 with maximum capacity of 192 megawatts per hour through 4 gas turbines producing 16.2 million gallons of water daily through three distilling units. In 2001, AMPC water production increased to 38.7 million gallons daily by adding three extra distilling units in Mirfa. In 2008 Al Ain Power Plant, with a 256 megawatts production, joined AMPC. The total current production of AMPC is 551 megawatts and 38.7 million gallons of water daily.





3 Environment Agency – Abu Dhabi

The Environment Agency – Abu Dhabi (EAD) was created from law No. 4 of 1996 pertaining to the establishment and development of the Environmental Research and Wildlife Development Agency. EAD is committed to protecting and enhancing air quality, groundwater as well as the biodiversity of our desert and marine ecosystem. By partnering with other government entities, the private sector, NGOs and global environmental agencies, we embrace international best practice, innovation and hard work to institute effective policy measures. We seek to raise environmental awareness, facilitate sustainable development and ensure environmental issues remain one of the top priorities of our national agenda.

Since 2008, EAD has been awarded ISO 9001 for Quality, ISO 14001 for Environment, ISO18001 for Health and Safety and ISO 22301 for Business Continuity Management System.

OUR VISION:

‘Towards a sustainable environment for a sustainable future’.

OUR MISSION:

‘To protect and conserve the environment for people’s wellbeing and a better life for all.’

OUR VALUES:

- Integrity
- Initiative with Innovation
- Empowerment
- Ownership and Accountability
- Excellence

Our main task is to ensure that the phenomenal growth of Abu Dhabi is achieved in a way that ensures a sustainable future for our emirate, the well-being of its residents and a good quality of life for all, now and in the future.

In pursuing this challenge since 1996, we have amassed a wealth of experience and passed some significant milestones. To name just a few examples, we have developed a comprehensive enforcement framework which includes a range of tools such as inspections and prosecutions and we upgraded our environmental permitting system to world’s best practice. We also gained a deep knowledge

of our fisheries, helped bring the Arabian Oryx back from the brink of extinction, preserved the world's densest dugong population and achieved a substantial increase in general public awareness of environmental issues.

Yet the challenges still remain in large numbers, as does our prime responsibility for the environmental quality of Abu Dhabi, and our duty to uphold and enforce the nation's environmental regulations.

To this end, we have a number of core elements that are at the heart of all we do:

We value and are committed to creating and maintaining effective partnerships with local, regional, national and international organizations which have been key to our success.

Our dedicated and committed employees are our greatest assets in fulfilling our mission.

We value deliberative, science-based decision making in the development of environmental policies.

We learn from best practices and we make our own best practices available to others.

We value and are committed to creating and maintaining effective partnerships with local, regional, national and international organizations which have been key to our success.



4 Abu Dhabi Education Council (ADEC)

Abu Dhabi Education Council (ADEC) was established in accordance with law No. 24 of 2005, issued by His Highness Sheikh Khalifa Bin Zayed Al-Nahyan, the UAE President, the Supreme Commander of the Armed Forces and the Ruler of Abu Dhabi. The Chairman of ADEC is H.H. General Sheikh Mohammed Bin Zayed Al Nahyan Crown Prince of Abu Dhabi and Deputy Supreme Commander of the UAE Armed Forces and the Vice-Chairman is H.H. Sheikh Mansour Bin Zayed Al-Nahyan, Deputy Prime Minister and Minister of Presidential Affairs to overlook the educational sector in the Emirate of Abu Dhabi. ADEC endeavors to transform the emirate into an innovation-based, knowledge producing society which is strongly connected to the world's global society and economy, yet deeply rooted in its culture and heritage.





مجلس أبوظبي للتخطيط العمراني
ABU DHABI URBAN PLANNING COUNCIL

5 Abu Dhabi Urban Planning Council

The Abu Dhabi Urban Planning Council (UPC) was created by Law Number 23 in the year 2007 as the agency responsible for the future of Abu Dhabi's urban environments and as the expert authority behind the visionary Plan Abu Dhabi 2030: Urban Structure Framework Plan published in September 2007.

The Abu Dhabi Urban Planning Council initiates, drives and supports Abu Dhabi's urban development strategy. By transferring the Plan Abu Dhabi 2030 vision to physical settings, the UPC will develop strategic development plans that create guiding principles which direct critical projects designed to shape the Emirate. The vision is built on a comprehensive analysis of the urban fabric, land availability and its best use, environmental issues, mobility, infrastructure and urban services that need to be integrated in to the overall development strategy as a partnership between the government, private development interests, and its citizens. With the expertise of the UPC, the Abu Dhabi Government will ensure the development of a professionally designed and well-managed urban environment in the Emirate's towns and cities, complete with world-class traffic and transport systems.

The government of Abu Dhabi has directly committed itself to strengthen and develop four key priority areas:

- Economic development;
- Social and human resources development;
- Infrastructure development and; and
- Environmental Spatial Systems.

The simultaneous development of the Regions is to keep pace with that of the Capital. It is also an important policy priority to achieve an Emirate-wide distribution of economic activities and associated benefits. The Abu Dhabi 2030 Urban Structure Framework Plan, which offers a vision for the evolution of the city over the next 15 years, provides a blueprint for Abu Dhabi's long-term success. As the first planning program of this kind and scope within the region, the work of the UPC will become a best-practice benchmark for future urban design within the UAE and beyond.

To help achieve the aforementioned goals, part of the UPC's work is to produce plans, policies and develop regulations that govern Abu Dhabi's physical environment, in line with the targets identified by the Abu Dhabi Vision 2030. In addition to this, the UPC is entrusted with the management of Abu Dhabi's sustainable urban growth through visionary planning, efficient development review and proactive implementation in coordination with the public and private sector partners.

Within this scope, the UPC's planning and Infrastructure Sector is responsible for the following:

- Prepare the Sector Strategy to align with the UPC 5-Year Strategic Plan;
- Ensure Sector and Department Manuals are up to date and align with leading practice standards;
- Prepare and test strategic 2030 Framework Plans, Regional Plans and Detailed Plans for all three regions of the Emirate of Abu Dhabi;
- Produce Master Plans for UAE National Housing;
- Prepare revitalization plans to improve existing communities;
- Explore the design and feasibility of visions, concepts and projects from the Government;
- Develop policies, manuals, guidelines and regulations for the Emirate of Abu Dhabi;
- Study and recommend appropriate allocation of land for projects within the Emirate of Abu Dhabi;
- Prepare Implementation Plans;
- Receive and incorporate input into plans from all the UPC core business departments;
- Collaborate and liaise with various Government Departments and stakeholders; and
- Maintain specialized tools (e.g. Geo planner), to allow for user-friendly access to all GIS data to assist in Plan development.

In the UPC, the Planning and Infrastructure Sector is a pivotal department as it contributes to the development of a solid background for all current and future urban developments in the Emirates of Abu Dhabi.



6 Regulation and Supervision Bureau

The Regulation and Supervision Bureau (RSB) is the independent regulatory body for the water, wastewater and electricity sectors in the Emirate of Abu Dhabi. The RSB was established in terms of Law No. 2 of 1998, as amended concerning the Regulation of the Water and Electricity Sector.

The Abu Dhabi water, wastewater and electricity sectors are governed by Law No. 2 of 1998 as amended and by Law No. 17 of 2005, as amended concerning the Establishment of Abu Dhabi Sewerage Services Company (ADSSC).

The primary duty of the RSB is to ensure the continued availability of potable water for human consumption as well as electricity for use in hospitals and centres for the disabled, aged and sick in the Emirate of Abu Dhabi. General duties encompass a



wide range of issues including the protection of customers and the development of safe and reliable water, wastewater and electricity networks.

Particular responsibilities include:

- Ensuring the security of the supply of water, electricity and wastewater services in the Emirate of Abu Dhabi;
- Ensuring the supply of water and electricity and connection to sewerage networks to all consumers;
- Ensuring the availability of health and safety guidance regarding water, wastewater services and electricity supply to the public;
- Publishing information relating to standards of performance by licensed operators;
- Taking account of national and international environmental standards as they affect the water, wastewater and electricity sector and consult with relevant bodies in the Emirate when necessary in relation to the interests of the consumer and sector;
- Having special regard to the interests of those persons whose lives may be endangered by the lack of potable water, electricity or sewerage services and others with special needs in connection with the cost and method of supply and the use of appliances and fittings;
- Promoting competition in the water, wastewater and electricity sector;
- Ensuring the operation and development of a safe, efficient and economic sector in the Emirate of Abu Dhabi; and
- Protecting the interests of consumers as to the terms and conditions and price of supply.
- Protecting the interests of users of sewerage services in the Emirate.

In essence, the RSB enforces the relevant laws through the licensing of activities to various entities who undertake a regulated activity in the sector. Regulated activities include: generation, transmission, distribution and sale of electricity and the production, transmission, distribution, sale and treatment of water and the collection, treatment and disposal of wastewater.

Once a licence is issued, the RSB will monitor, modify, where needed, and enforce conditions in licences to participant companies. It also has the power to establish and monitor technical, performance, safety and customer standards. Approvals from RSB must be obtained if licensed operators wish to dispose of any or all of their assets, or purchase the assets of another generator, transmitter or distributor.

Internal Requirements of the Regulation and Supervision Bureau

The following documents are RSB guidance documents for infrastructure development and licensing requirements in the Emirate of Abu Dhabi which are published on the RSB's website.

- RSB, Guide for Major Developers and New Entrants; and
- RSB, Guide to the Bureau's Licensing Process for New Entrants.

Infrastructure should also comply with the following sector regulations and approved codes which are published on the RSB's website:

- Electricity Distribution Code
- Electricity Transmission Code
- Metering and Data Exchange Code
- RSB, The Customer Metering Regulations
- RSB, The Electricity Supply Regulations
- RSB, The Electricity Wiring Regulations
- RSB, The Recycled Water and Bio solids Regulations
- RSB, The Trade Effluent Control Regulations
- RSB, The Water Quality Regulations
- RSB, The Water Supply Regulations
- Water Distribution Code
- Water Transmission Code



7 Center of Waste Management (Tadweer)

Established in December 2008, The Center of Waste Management - Abu Dhabi (Tadweer) is the lead agency responsible for controlling and coordinating waste management activities throughout the Emirate of Abu Dhabi. This includes coordinating the policy, strategy, and contractual systems of waste management, as well as implementing the Waste Management Strategy and all aspects of service delivery required to establish a full-cycle integrated waste management system.

Furthermore, the key objectives of the CWM (Tadweer) are to:

1. Developing necessary plans to modernize and develop waste management operation in sectors and entities intended in the emirate.
2. Developing studies; preparing the documents of technical tenders announcing such; and qualifying private companies to deal with wastes in the emirate.
3. Managing projects related to collecting, removing, treatment and disposal of wastes administratively, financially and operatively with entities related.
4. Assigning consultant and legal companies to review and evaluate contracts concluded on collecting, moving and treatment of wastes and developing technical standards of operation and follow up.



5. Designing and implementing projects using BOT system to deal with hazardous and non-hazardous wastes in the emirate.
6. Managing facilities of waste management projects such as factories of fertilizers; waste disposal areas; and sort stations, collection, incineration and other waste treatment and disposal stations.
7. Producing a database for wastes produced; making and developing an electronic system to trace wastes from source to disposal areas; performing scientific studies suitable for putting strategic plans in the emirate.
8. Developing and implementing educational programs related to minimizing the production, reusing and sorting of wastes.
9. Training national technical cadres who work in the field of wastes.
10. Revising and approving operation plans, maintenance and environmental emergencies for establishments of storage, treatment and disposal of wastes.
11. Coordinating with competent authority to develop the provisions of Law No (21) of the year 2005 on waste management in the Emirate of Abu Dhabi.
12. Any other specializations assigned to the Center by competent authority in the field of hazardous (excluding nuclear waste), non-hazardous, solid and liquid (excluding sewage) waste management in the emirate.

8 Abu Dhabi Sewerage Services Company



شركة أبوظبي لخدمات الصرف الصحي
Abu Dhabi Sewerage Services Company

8.1 *Mandate of the Abu Dhabi Sewerage Services Company*

ADSSC was established in terms of Law No. 17 of 2005 Concerning the Establishment of Abu Dhabi Sewerage Services Company. ADSSC owns, maintains and operates all public domain sewerage infrastructure, including wastewater treatment plants, sewerage collection networks and recycled water transmission mains in the Emirate of Abu Dhabi.

ADSSC is responsible for the collection, treatment and disposal of wastewater discharged from all residential, commercial and industrial customers in the Emirate of Abu Dhabi including the safe disposal of both the solid and liquid residuals (bio solids and recycled water).

ADSSC core business processes address the following 3-activities:

- Waste water collection
- Waste water treatment
- Waste water disposal

ADSSC provides treated water and bio-solids to the Municipalities for horticulture purposes. Over 60% of the recycled water is used for irrigation purposes, thereby, contributing towards environmental preservation. With the completion of treated water delivery pipelines, in the near future, 100% of treated wastewater produced by ADSSC will be reused. ADSSC contributes effectively to the Emirate of Abu Dhabi vision and is a key link in the overall value-proposition offered to private, public and commercial customers. ADSSC services form an integral part of the Government of Abu Dhabi utilities value-chain. ADSSC endeavours to ensure that its services remains invisible, safe, odour-less and environmentally-friendly.

Today, ADSSC is a pioneer in developing the infrastructure of Abu Dhabi and its projects are known worldwide. ADSSC attracts global players with construction and design expertise and together we utilize advanced technologies in all business operations with the Strategic Tunnel Enhancement Programme (STEP) and its Wastewater Treatment Plants being good examples. ADSSC has been recognized for its work with an accolade of international awards in many different work areas

ADSSC has gained an international recognition for Best Practices by the Australian Water and Waste Water Benchmarking Association in 2008. ADSSC was also bestowed Sewerage Master Plan award, at the BEX Conference in Valencia, Spain, in 2009. ADSSC was the first UAE business to achieve triple ISO Certifications (ISO 9001, ISO 14001 and OHSAS 18001) in 2010. ADSSC has also achieved the “Silver Award for built projects category at the International Awards for Liveable Communities”, and the “Golden Employer” Classification in Abu Dhabi TAWTEEN Council website. ADSSC has also achieved 3 international awards in 2012 and 2013 for GIS implementations. The Balanced Scorecard Hall of Fame Award for Executing Strategies was given in 2013. ADSSC was the first government entity in Abu Dhabi Emirate to be awarded in the Utilities and Energy Category.

8.2 *Internal Requirements of the Abu Dhabi Sewerage Services Company*

The following documents contain the internal requirements and procedures for infrastructure provision under the jurisdiction of the ADSSC:

- ADSSC’s Integrated Management System (IMS);
- ADSSC, applicable regulations and guidelines (Technical Specifications and Design Guides)





9 Abu Dhabi General Services Company (Musnada)

Abu Dhabi General Services “Musnada” was established in Dec 2007 by Law 27/2009 as a Public Joint Stock Company to provide support services to government entities in the areas of facilities management, construction, contract management, IT, Catering, Logistics, Stores etc. Our current offering focuses on Design & Construction in addition to Facility Management. Musnada supports the delivery of Abu Dhabi Economic and Urban Planning visions

Vision

- Page Content
- Leading the Abu Dhabi’s infrastructure ambition.

Mission

- Page Content
- Provide integrated, cost effective and sustainable delivery and maintenance of Abu Dhabi government Assets.



10 ADNOC Distribution

Founded in 1973, as the first company owned by Abu Dhabi Government specialised in the marketing and distribution of petroleum products in the UAE, Abu Dhabi National Oil Company for Distribution (ADNOC Distribution) is a major driver of the country’s economy and an important community player. It employs over 14,000 people of 50 different nationalities from all over the world. ADNOC Distribution is part of ADNOC Group – ranked among the top ten energy companies worldwide.

ADNOC Distribution’s customers can fuel, shop, have a quick oil change or get their car washed at the company’s wide network of service stations. ADNOC Distribution offers car care through Autoserv, inspections via our Vehicle Inspection Centres and supplies LPG cylinders to the UAE’s households. The company

meets the daily energy needs of the country's industries, while providing fuel and lubricant products for aviation, marine and fleets.

Throughout the GCC, the wider Middle East, Far East, Africa, and the CIS countries, ADNOC Distribution markets lubricants under the sub-brand ADNOC Voyager, which exceeds the highest international specifications.

ADNOC Distribution is also developing the natural gas (NG) downstream business in Abu Dhabi, catering to the requirements of the residential, commercial, transportation, small and medium industrial, and district cooling segments. ADNOC Distribution currently manages NG networks in Yas Island, Al Raha Beach and Al Maryah Island. In addition, the company operates compressed natural gas (CNG) refilling stations and conversion centres, serving an ever-increasing customer base keen to make the switch to alternative green fuels.



11 Ministry of Interior (Abu Dhabi Police and Civil Defence)

The Ministry of Interior is responsible for traffic control and civil defence services on a federal basis. The General Command of the Abu Dhabi Police, under the auspices of the Ministry of Interior (MOI), is responsible for and has an interest in Infrastructure services from a law enforcement and public safety perspective. In particular the Abu Dhabi Police has an interest in road infrastructure with regard to:

- Traffic control;
- Enforcement of traffic rules and regulations;
- Crash mitigation and prevention; and
- Emergency response to incidents.

The General Directorate of Abu Dhabi Civil Defence (GDCCD) has a fire safety interest in infrastructure systems. The GDCCD works closely with the DMA and other authorities to ensure the safety of infrastructure in the various communities. The above authorities are primarily governed by federal laws and regulations.



12 Telecommunications Regulatory Authority

The Telecommunications Regulatory Authority (TRA) of the United Arab Emirates (UAE) has been established according to the UAE Federal Law by Decree No. 3 of 2003 – Telecom Law. The TRA is mandated to manage every aspect of the telecommunications and information technology industries in the UAE. It is entrusted with a wide range of responsibilities related to the Telecommunications and Information Technology Sector, both within and outside the UAE. These responsibilities include:

Ensure that the telecommunications services provided throughout the state are sufficient to satisfy the public demands of those who wish to make use of such services

Enhance the level of service provided by the telecommunications sector in order to promote the interests of such services

Ensure that licensees meet quality standards of performance and adhere to the terms and conditions of the licenses granted to them

Encourage, promote, and develop the telecommunications and information technology industries in the state

Promote and enhance the telecommunications system in the state as indicated by the development and the establishment of industry related training institutions and through the availability of the latest apparatus, equipment, and facilities provided by telecommunications technologies

The TRA's vision is to achieve a global leadership position in ICT sector. Its mission is to strive to be a leading organization in the ICT sector in United Arab Emirates, committed to maintaining positive competition to protect the interests of the subscribers, and promoting electronic transformation of the federal agencies and their services, by relying on national competencies to apply the best international standards and practices in supervision of the sector's and to encourage innovation and investment.

On the long term, the TRA aims to achieve the following strategic goals:

Regulating the Telecommunications Sector in the UAE and enforce a regulatory framework that stimulates Competition and elevating the level of quality of the provided services

Striving to ensure the provision of the required resources in a fair manner to develop the best services in the ICT sector

Establishing the bases for the e-government at the federal level

Promoting the electronic lifestyle in the UAE

Ensuring that all administrative services are in accordance with the standards of quality, efficiency and transparency

Consolidate Innovative Culture within the working environment

13 National Central Cooling Comapany (Tabreed)



13.1 *National Central Cooling Company (Tabreed)*

Tabreed is a UAE-based utility company that provides energy-efficient, cost-effective and environmentally-friendlier year-round district cooling solutions in the GCC. Founded in 1998, and listed on the Dubai Financial Market, Tabreed's cooling infrastructure is an integral part of the region's growth. The company now delivers 1,011,900 Refrigeration Tons (RT) to major residential, commercial, government and private projects. Tabreed owns and operates 70 plants in its portfolio across the GCC, including 62 plants in the United Arab Emirates, three plants in Qatar, two in the Kingdom of Saudi Arabia, two in Oman, and one in the Kingdom of Bahrain.

Tabreed provides its district cooling services to many of the region's landmark projects including all the developments on Abu Dhabi's Al Maryah Island, home to Cleveland Clinic Abu Dhabi and Abu Dhabi Global Market, and all the developments on Yas Island such as Ferrari World, Yas Marina Circuit and Yas Mall, in addition to other national and regional landmarks including Sheikh Zayed Grand Mosque, Dubai Metro, Dubai Parks & Resorts, Pearl Island in Qatar, and Aramco Development Project in Al Dhahran in the Kingdom of Saudi Arabia.



13.2 *Our Vision, Mission and Core Values*

Our Vision: “To be the leading cooling provider by utilizing sustainable, reliable & cost efficient energy solutions.”

Our Mission: “Creating value through optimizing, innovating and striving for operational excellence to exceed stakeholder expectations, whilst protecting people, assets and environment.”

Our Core Values:

- Inspire with Positive Energy
- Dedicated to Excellence
- Stronger Together
- Resilient in the face of Challenges
- Committed to Health, Safety and the Environment

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